The Conditional Activation Model: Uncovering Policy Demand Responses to AI-Driven Automation

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Abstract

Why do citizens often fail to update their policy preferences when presented with objective evidence of economic threats? We address this "information inertia" puzzle through the lens of AI-driven job automation. While standard accounts suggest that information about personal automation risk should increase demand for government protection, recent experiments frequently report null effects. We propose the Conditional Activation Model (CAM), a "funnel of causality" framework positing that information activates policy demand only when it passes two sequential gates: (1) source trust and (2) dispositional vigilance (anxiety) triggered by a positive shock to prior beliefs. Leveraging a pre-registered experiment with 901 U.S. workers and a Bayesian finite mixture model to recover latent trust classes, we demonstrate that standard Average Treatment Effects (ATE) mask deep persuasion. While the aggregate effect is modest, information increases policy demand by 0.61 standard deviations among the theoretical subgroup of high-anxiety "source Trusters" who learn they face greater risk than expected. These results reconcile conflicting findings in the literature and challenge the utility of the unconditional ATE for studying how citizens respond to economic threats.

1 Automation Risk, Economic Insecurity, and Demand for Government Protection

A central puzzle in political behavior is the "information inertia" of the electorate: despite being exposed to expert estimates of economic threats, citizens do not always update their policy preferences. The rapid emergence of Artificial Intelligence (AI) and job automation risk provides a test case for understanding this phenomenon. One might expect that informing individuals about their automation risk would increase support for government intervention. However, empirical evidence shows heterogeneous and null effects, suggesting that the relationship between risk information and policy preferences is conditional rather than direct. Existing studies reach mixed conclusions: some find that automation risk mobilizes support for government intervention (Golin and Rauh, 2022; Wu, 2023), others report null effects where information fails to move policy preferences (Zhang, 2022; Heinrich and Witko, 2024), and still others show that responses are highly context-dependent (Gallego and Kurer, 2022; Jeffrey, 2023).

To make sense of these inconsistencies, this paper proposes a **Conditional Activation Model** (or "Funnel of Causality") of policy demand. We argue that providing objective facts about automation risk is necessary but insufficient to change policy preferences. Instead, the causal effect of expert information is conditional on two sequential gates:

1. **Gate 1 (Trust)**: The subject must trust the source of the information (e.g., experts' estimates of job automation risk). If they do not, the message is rejected, and the treatment effect is diminished.
2. **Gate 2 (Activation Potential and Disposition)**: Among those who trust the source, the message only activates economic anxiety and increased downstream policy demand if there is sufficient "potential for activation" — experts' risk estimate is higher than the individual's prior belief, i.e., a positive "Belief Gap" — and the individual has the psychological disposition to react to it (Dispositional anxiety).

Hence, the effect of risk information on policy preferences should be strongest among individuals who trust experts, face a positive Belief Gap, and exhibit high trait anxiety. Conversely, individuals who distrust experts or whose prior beliefs align with or exceed expert estimates are unlikely to update their preferences.

First, we expect heterogeneous treatment effects based on **trust in experts**. Trust in scientific and expert institutions has declined in recent decades, leading to increased skepticism toward expert information (Gauchat, 2012; Pew Research Center, 2024). Individuals who distrust experts are less likely to fully accept expert risk information (Chen, Fraser and Cuddeback, 2018), which should attenuate—though not necessarily eliminate—treatment effects for this group. Even Skeptics may exhibit modest policy responses. Priming effects draw attention to automation as a salient issue, while framing effects signal that automation risk merits consideration. Thus, we hypothesize a *gradient* in treatment effects between "Skeptics" (low trust) and "Trusters" (high trust), with Trusters exhibiting larger responses.

Drawing on Affective Intelligence Theory, we conceptualize trait anxiety as a dispositional tendency toward vigilance that activates accuracy-motivated processing. High anxiety tends to shift individuals away from habitual reliance on partisan or ideological heuristics and toward more effortful information evaluation (Marcus, Neuman and MacKuen, 2000; Valentino et al., 2008; Albertson and Gadarian, 2015). We therefore expect that risk information will have the strongest effects among individuals who both trust the source and possess high trait anxiety.

This theoretical framework implies that a standard Average Treatment Effect (ATE) is **the wrong estimand for our purposes**: it dilutes the signal by averaging over "Skeptics" (who are predicted to show attenuated responses) and low-disposition individuals, and it obscures the conditional nature of activation among "Trusters." While we may observe a positive average effect driven by priming and framing, the theoretically appropriate quantity is a **Conditional Average Treatment Effect (CATE)**—the effect among respondents who pass both gates. This estimand captures the larger response that the unconditional average masks.

Hence, to address these challenges, we propose a Bayesian finite mixture model that jointly estimates (1) latent trust classes (Skeptics vs. Trusters) independently of expert information and (2) heterogeneous treatment effects conditional on trust and anxiety levels. This approach allows us to estimate the theoretically relevant CATE while modeling trust as a latent pre-treatment class and accounting for the psychological disposition of individuals.

Our pre-registered experiment with $N = 910$ U.S. workers yields three main findings. First, the unconditional ATE of personalized automation risk information on policy demand is positive and consistent across OLS, SEM, and the Bayesian mixture model ($d \approx 0.23$–0.28), confirming that expert information shifts preferences on average. Second, this aggregate estimate masks the heterogeneity that CAM predicts: approximately 62% of respondents are classified as Skeptics whose average treatment effects are indistinguishable from zero, while the maximum conditional effect among high-anxiety Trusters with positive Belief Gaps reaches $d = 0.61$ (95% CI: [0.40, 0.86])—more than double the unconditional ATE. Third, an activation threshold emerges at approximately the 20th percentile of trait anxiety, below which even trusted, expectation-violating information fails to generate policy demand. These results contribute a two-gate framework that organizes the mixed findings in the automation literature as a composition effect—heterogeneous subgroups canceling in aggregation—rather than genuine preference stability, and they identify a collider structure in post-treatment trust measurement that requires structural modeling rather than naive sample splitting.

1.1 Empirical Puzzle: Heterogeneous and Null Effects in Automation Experiments

Recent experimental studies on automation and policy preferences, however, present a puzzle that motivates our theoretical framework. Zhang (2022) conducted three experiments and reports: "Although informational treatments were effective at changing respondents' **beliefs about the future of work**, they **did not affect their policy preferences**." Manipulation checks confirmed that participants received and processed the information—yet policy demand remained unchanged.

This belief-preference disconnect is not an artifact of measurement design. Magistro et al. (2024) explicitly tested whether the null effects resulted from using generic policy items that failed to activate automation-specific con-

cerns. Their study employed *fully contextualized* items—"government policies designed to mitigate the negative consequences of automation," including an automation tax—across four countries (Australia, Canada, UK, US). Despite this domain-specific framing, they found "no discernible differences across treatment groups," leading them to conclude that "policy preferences are relatively fixed." Ladreit (2022) similarly framed policies around "workers who lose their jobs" to automation and found null effects despite increased automation worry.

Yet there is evidence of heterogeneity beneath these null averages. Arntz, Gregory and Zierahn (2024) examined treatment effects conditional on prior beliefs about automation and found "**heterogeneous and opposing treatment effects** on policy demand. Technological optimists respond to the treatment by **reducing demand** and support for policy interventions, whereas we find somewhat weaker, **opposite responses for neutral respondents and none for pessimists**." The result: null aggregate effects masking real persuasion in opposite directions. This is the first direct evidence that null ATEs in the automation literature may reflect *canceling effects* rather than genuine preference stability.

Additionally, Heinrich and Witko (2024) found policy effects only among respondents with high prior technology knowledge—a potential proxy for trust in expert forecasts, though the construct confounds multiple mechanisms. These findings suggest that null aggregate effects may partly reflect unmodeled heterogeneity in how individuals respond to expert information.

We propose that information may be received and processed (as evidenced by belief change in manipulation checks) while policy updating remains conditional on source acceptance. Standard manipulation checks confirm message exposure but do not measure whether participants *accept* the source as credible. If trust moderates treatment effects—with some individuals updating policy preferences while others discount the information—then pooling across these groups yields attenuated aggregate estimates. Our approach differs from prior work by explicitly modeling trust as a latent class, allowing us to estimate heterogeneous treatment effects by source acceptance.

2 Theory: The Conditional Activation Model

This section develops the Conditional Activation Model (CAM), a two-gate framework in which expert information about automation risk must clear two sequential conditions before it translates into policy demand. At *Gate 1*, source trust filters whether the information is accepted or dismissed: individuals who distrust the expert source discount the message regardless of its content. At *Gate 2*, a joint condition—the interaction between the Belief Gap (the discrepancy between expert and prior risk assessments) and trait anxiety—determines whether accepted information activates policy search among those who cleared the first gate. This framework provides a theoretical account of why recent automation experiments report a mixed pattern of null aggregate effects alongside evidence of conditional persuasion when heterogeneity is explicitly modeled.

2.1 Gate 1: Source Trust as a Necessary Condition

A foundational insight from the political persuasion literature is that source credibility functions as a cognitive shortcut: when individuals trust the source of a message, they are more likely to accept its content; when they distrust the source, the same information is discounted or rejected (Lupia and McCubbins, 1998; Zaller, 1992). Druckman (2001) demonstrates that source credibility moderates framing effects, and subsequent work documents systematic partisan asymmetries in trust toward scientific experts (Gauchat, 2012; Motta, 2018). If source trust matters for how individuals process expert information, then the null aggregate treatment effects reported by several automation experiments may reflect unmodeled heterogeneity in source acceptance rather than genuine preference stability.

Recent experimental evidence supports this interpretation. Zhang (2022) conducted three experiments and reports that "informational treatments were effective at changing respondents' **beliefs about the future of work**" but "**did not affect their policy preferences**"—manipulation checks confirmed message receipt, yet policy demand remained unchanged. Magistro et al. (2024), varying infoGap Slope Steepens With Anxiety Anxiety Slope

Steepens With Belief Gap rmation about automation's trade-offs across four countries, found that while fairness perceptions shifted, policy preferences showed "no discernible differences across treatment groups"—even though their policy items were explicitly framed as responses to automation. Ladreit (2022) similarly found null policy effects despite increased automation worry. The standard reading is that policy preferences are resistant to job automation information. Yet Arntz, Gregory and Zierahn (2024), examining treatment effects conditional on prior beliefs, found "**heterogeneous and opposing treatment effects**"—technological optimists *reduced* policy demand while neutral respondents increased it, yielding null effects only in aggregate. Heinrich and Witko (2024) found policy effects exclusively among respondents with high prior technology knowledge, a construct likely correlated with trust in expert forecasts.[1] Shen et al. (2025), in a field experiment with Chinese white-collar workers, found that generative-AI information increased support for progressive social policies—but effects were concentrated among prior AI users, again suggesting that baseline familiarity with the technology (and plausibly trust in forecasts about it) conditions responsiveness. These patterns are consistent with a trust-moderated account: pooling across individuals who accept versus reject the expert source masks genuine heterogeneity beneath null averages.

A further complication is that trust in expert sources is not exogenous to the information treatment. When an expert forecast sharply contradicts an individual's prior beliefs, the resulting cognitive dissonance can be resolved either by updating beliefs toward the expert or by discounting the source itself (Taber and Lodge, 2006). Most studies in this literature cannot diagnose this mechanism because they do not elicit prior beliefs. This source-rejection pathway may be especially pronounced in the automation domain, where risk information directly threatens economic identity and self-concept as a competent, secure worker—activating identity-protective cognition (Kahan et al., 2007; Kahan, 2017) and directional reasoning (Druckman et al., 2022). Im et al. (2019) document that automation anxiety varies systematically with individual economic security, and Thewissen and Rueda (2019) find observational links between automation exposure and political attitudes, suggesting the threat is psychologically potent. Coppock (2023) finds that persuasion often operates "in parallel" in standard policy domains, but this finding rests on cue-free experimental designs and does not rule out motivated reasoning—as Druckman et al. (2022) and others note, directional goals are a primary reason persuasion efforts fail, and automation risk, with its direct identity relevance, is precisely the kind of domain where source acceptance should vary across individuals. Our design addresses the endogeneity problem directly: by eliciting respondents' subjective automation risk *before* presenting the expert forecast, we can compute each individual's Belief Gap and test whether trust in the expert source varies as a function of this discrepancy.[2] Whether null effects reflect genuine preference stability or source rejection has radically different implications: under stability, information campaigns are futile regardless of design; under source rejection, more carefully calibrated interventions that account for prior beliefs might succeed.

2.2 Gate 2: Activation Potential and Psychological Disposition

Even among individuals who trust the expert source, accepting information about automation risk does not automatically generate policy demand. Affective Intelligence Theory (AIT) offers a mechanism for when accepted information triggers deeper processing: the brain's surveillance system monitors the environment for novel threats, and when it detects a discrepancy between expectations and incoming information, anxiety activates a shift from habitual reliance on standing dispositions toward deliberative, accuracy-motivated reasoning (Marcus, Neuman and MacKuen, 2000; Valentino et al., 2008). Crucially, Albertson and Gadarian (2015) show that anxious citizens do not simply abandon all heuristics—they increase their reliance on *trusted* experts, linking the activation mechanism of Gate 2 back to the trust filter of Gate 1. We distinguish here between *state anxiety*—the transient emotional response that a threatening message may evoke—and *trait anxiety*, a stable dispositional tendency toward vigilance and threat-sensitivity that, under AIT, governs the threshold at which the surveillance system engages deliberative processing (Marcus, Neuman and MacKuen, 2000). We operationalize

[1] Technology knowledge may capture familiarity with AI capabilities, general cognitive ability, or differential exposure to automation discourse, in addition to trust in expert sources.

[2] We document an inverted U-shaped relationship between the Belief Gap and source trust: trust is highest when expert estimates roughly match prior beliefs (Gap ≈ 0) and declines as the discrepancy grows in either direction (see Appendix, Table 9). This "epistemic validation" pattern is consistent with motivated reasoning accounts (Taber and Lodge, 2006).

the informational trigger as the *Belief Gap*: the difference between the expert forecast of automation risk and the respondent's own prior assessment (Expert − Subjective). A positive Belief Gap signals that the expert sees greater risk than the individual expected—precisely the kind of expectation-violating "news" that AIT predicts will engage the surveillance system. Yet the magnitude of this informational shock alone may be insufficient: trait anxiety determines whether the surveillance system sustains deliberative processing long enough to revise policy preferences or whether the individual reverts to heuristic dismissal. The interaction between the Belief Gap and trait anxiety is therefore the core theoretical quantity at Gate 2—it captures not merely whether individuals received threatening information, but whether they possess both the informational stimulus and the dispositional capacity to translate that stimulus into policy search.

The distinction between informational shock and policy updating also helps explain a recurring puzzle in the automation literature: treatments that successfully increase anxiety without shifting policy preferences. Zhang (2022) found that respondents updated beliefs about the future of work but not policy demand; Ladreit (2022) similarly observed heightened automation worry with no downstream policy effects. Druckman et al. (2022) draw a useful distinction between shallow priming—in which a treatment temporarily raises the salience of an issue—and deep persuasion, which requires sustained elaboration sufficient to revise durable preferences. Brief informational vignettes, the dominant design in this literature, may accomplish the former without the latter: respondents acknowledge the threat (hence positive manipulation checks) but do not engage in the effortful processing needed to connect that threat to specific policy positions. This suggests at least three distinct pathways to null aggregate effects: source rejection (Gate 1 failure), shallow priming without elaboration (Gate 2 failure despite message acceptance), and deep processing that nonetheless produces motivated resistance among those whose identities are threatened. Our experimental design addresses the engagement-depth concern through a three-step protocol—eliciting prior beliefs, presenting personalized expert feedback, and asking respondents to evaluate the discrepancy—that requires active elaboration rather than passive receipt. Haslberger (2024) provides convergent evidence: a behavioral treatment in which respondents actually used an AI system to perform work tasks produced significant policy effects, consistent with the hypothesis that deeper engagement with automation's implications is necessary to shift preferences. Positive manipulation checks, in this light, may capture salience rather than the kind of belief revision that precedes preference change.

An important refinement is that the Gap × Anxiety interaction activates policy *search*, not policy *direction*. Recent work shows that automation threat channels into qualitatively different policy responses depending on prior dispositions: Gallego and Kurer (2022) find that automation-threatened workers favor protectionist measures over redistribution, Wu (2023) document partisan divergence in preferred response type, Jeffrey (2023) show that affective framing shifts which specific policies gain support, and Lee (2024) finds that priming distributive consequences of automation (who gains, who loses) shapes whether respondents favor restriction or compensation—with institutional context (UK vs. Denmark) further moderating the direction. CAM accommodates this heterogeneity by treating ideology as a downstream filter: the two gates determine *whether* individuals enter a state of active policy search, while pre-existing ideological commitments shape *what* they search for. Because our policy battery emphasizes government-led responses—retraining programs, social spending, regulation—it captures demands that align more naturally with left-leaning preferences, and we should expect the strongest treatment effects among left-leaning, high-anxiety Trusters with positive Belief Gaps.

A related consideration is whether policy demand reflects self-interested responses to personal job threat or sociotropic concern for affected workers more broadly. Haslberger (2024) finds stronger evidence for the sociotropic channel. CAM reframes this distinction: both pathways are downstream of Gate 1, because concern for others' automation risk—no less than concern for one's own—requires first accepting that the expert forecast is credible. An individual who dismisses the expert source has no basis for sociotropic worry. The sequential logic is thus: trust the source, accept the threat as real, extend concern to affected workers, and support protective policies.

2.3 Theoretical Implications: Why ATE May Be Insufficient

The two-gate structure yields a concrete decomposition of the sample into three theoretically distinct subgroups. *Skeptics* fail Gate 1: they discount the expert source and consequently exhibit attenuated or null treatment responses. *Trusters with negative or zero Belief Gaps* clear Gate 1 but fail Gate 2: they accept the source as credible yet learn that their risk is no higher than expected, producing no informational shock and minimal policy response. Only *Trusters with positive Gaps and high trait anxiety* pass both gates—they accept the expert, discover that risk exceeds their prior, and possess the dispositional capacity to sustain deliberative processing through to policy updating. When these three subgroups are pooled, the unconditional ATE averages strong effects among the third group against weak or null effects among the first two, diluting the signal (Druckman and Leeper, 2012). This is precisely the pattern that Arntz, Gregory and Zierahn (2024) observed empirically—opposing treatment effects that cancel in aggregate—but without a framework to explain *why* different individuals respond in opposite directions.

Table 1: Design Features and Findings Across Automation Policy Experiments

Study	**Trust Modeled?**	**Prior Beliefs Elicited?**	**Engagement Required?**	**Policy Framing**	**Finding**
Zhang (2022)	No	No	No (vignette)	Mixed	Null
Magistro et al. (2024)	No	No	No (conjoint)	Contextualized	Null
Ladreit (2022)	No	No	No (vignette)	Contextualized	Null
Arntz et al. (2024)	No	Yes (optimism)	No	Traditional	**Heterogeneous**
Heinrich & Witko (2024)	Proxy (knowledge)	No	No	Mixed	**Conditional+**
Haslberger (2024)	Reduced (behavioral)	No	Yes (AI use)	Traditional	**Positive**
This Study	**Yes (latent)**	**Yes (Gap)**	**Yes (3-step)**	**Contextualized**	**CATE**

Table 1 situates our design relative to prior automation experiments along the three dimensions CAM identifies as critical. Null ATEs cluster among studies that share one or more design features: no measurement of source trust, no elicitation of prior beliefs (precluding Belief Gap computation), and shallow treatment engagement (brief vignettes without elaboration requirements). No single prior study addresses all three. Our design innovates on each dimension—latent trust classification via a Bayesian mixture model, pre-treatment belief elicitation enabling individualized Belief Gaps, and a three-step protocol requiring active elaboration—creating the conditions under which the two-gate mechanism can be identified rather than obscured by aggregation.

2.4 Deriving Testable Predictions

The two-gate framework generates predictions at three levels: the aggregate effect, the trust gradient, and the conditional activation surface.

H1 (Weak Main Effect). Because the unconditional ATE pools Skeptics, low-Gap Trusters, and high-Gap high-anxiety Trusters, the aggregate effect will be positive but modest—genuine persuasion among the responsive subgroup is diluted by null or opposing responses elsewhere. A null or negative ATE would suggest the treatment is ineffective even among responsive subgroups; a large ATE ($d > 0.5$) would suggest heterogeneity is less severe than CAM posits.

H2 (Gate 1 Gradient). Treatment effects will be substantially larger among Trusters than Skeptics: $\tau_{\text{Truster}} > \tau_{\text{Skeptic}}$. Source rejection blocks information before it can activate policy search, so Skeptics should show attenuated responses at most. Equivalent effects across trust types would indicate that source acceptance does not moderate information processing in this domain.

H3 (Belief Gap Dose). Among Trusters, treatment effects will increase with the magnitude of the positive Belief Gap. A positive Gap provides the expectation-violating "news" that AIT's surveillance system requires; a zero or negative Gap provides no informational shock. If effects were constant across Gap values, updating would be categorical rather than dose-dependent.

H4 (Anxiety Amplification). Among Trusters with positive Gaps, high trait anxiety will amplify treatment effects.

Trait anxiety sustains deliberative processing; without it, the informational shock may dissipate before reaching policy preferences. No moderation by anxiety would suggest that AIT does not apply to economic threat processing, or that our trait measure fails to capture the relevant construct.

H5 (Conditional Activation). The largest treatment effects will occur among Trusters who possess high trait anxiety and receive positive Belief Gaps—the subgroup satisfying both gates. This cell of the CATE surface represents the strongest test of CAM's core prediction. A flat surface—similar effects across all Gap $\times$ Anxiety combinations among Trusters—would indicate that the two-gate structure is overspecified. If maximum effects occurred in a different cell (e.g., Low Gap $\times$ High Anxiety), it would suggest alternative mechanisms that CAM does not accommodate.

3 Research Design

3.1 Experimental Design and Sample

We conducted a pre-registered, between-subjects randomized experiment with $N = 901$ U.S. workers recruited via CloudResearch Connect. Eligibility required participants to be ages 18–65, currently employed (at least part-time), and English-speaking. We excluded full-time students, retirees, and respondents who failed embedded attention checks. Participants were randomly assigned to either the treatment condition (receiving personalized automation risk information) or a pure control condition.

3.2 Treatment: The Three-Step Belief Correction Procedure

Our treatment differs fundamentally from standard vignette-based experiments. Rather than presenting a single block of information about automation risk, we implement a *three-step sequential procedure* that creates individualized treatment intensity by anchoring respondents to their prior beliefs before revealing expert estimates:

Step 1: Expert Context. All treatment-group respondents first read a neutral explanation of how experts assess automation risk. The text describes the general logic of expert assessment: experts examine the component tasks of each occupation, identify which tasks are routine and clearly defined (and thus susceptible to automation), and note that jobs with more data-intensive, online, or repetitive workflows face higher risk. Importantly, this initial explanation does *not* yet reveal the specific source (Frey and Osborne) or the respondent's personal score—it merely educates respondents about the conceptual basis for expert assessments. A comprehension check ensures attention.

Step 2: Subjective Belief Elicitation. Before revealing any personalized information, we ask respondents to state their own belief about where their job falls on the automation risk distribution (0 = safest; 100 = riskiest). Crucially, this elicitation occurs *after* respondents understand the methodology but *before* they see expert data about their occupation. This sequencing creates psychological commitment to a prior estimate (cf. Festinger, 1957, on cognitive dissonance): respondents anchor on a self-generated belief, which amplifies the salience of any subsequent discrepancy. We also elicit subjective uncertainty by asking respondents "about how many positions above or below do you feel [your estimate] might be off from where experts would put your job," allowing us to construct a Bayesian measure of informational potential.

Step 3: Personalized Correction (The Contrast). Finally, we reveal the expert risk score for the respondent's specific occupation via a graphical bar-chart comparison. It is at this stage that we first attribute the source: respondents learn that "Dr. Carl Benedikt Frey and Dr. Michael A. Osborne are researchers at the University of Oxford" who developed the automation risk estimates. The visualization places the respondent's stated belief alongside the expert estimate on the same 0–100 scale, making the *Belief Gap* (Objective $-$ Subjective) visually concrete. This is not a framing manipulation—both groups receive identical information formats. Rather, the treatment intensity is the *magnitude and direction of the gap* between prior beliefs and expert data, which varies naturally across respondents.

Control Group. Respondents assigned to control proceed directly to outcome measures without receiving any

automation-related information, belief elicitation, or expert estimates. This clean control allows us to estimate the overall treatment effect of the belief-correction procedure.

This sequential structure distinguishes our design from prior work in three ways. First, by eliciting beliefs before revealing the expert estimate, we ensure that the information is processed as a *correction* to a committed prior rather than as novel information entering a blank slate, which should amplify updating among those who discover large gaps. Second, the Belief Gap provides a continuous measure of treatment dose, allowing us to model heterogeneous effects as a function of informational surprise rather than binary treatment assignment. Third, unlike studies that manipulate how information is presented (threat vs. opportunity framing), our procedure holds presentation constant: variation in treatment response arises from the *content* of the personalized feedback, not from experimentally manipulated framing.

3.3 Measurement

3.3.1 Primary Outcome: Policy Demand Index

Our primary outcome is a five-item index measuring support for government policy responses to AI-driven automation. Unlike prior studies that use generic redistribution items drawn from ANES or similar batteries, our policy items are *explicitly contextualized* as responses to automation's workforce impact. Respondents rated each item on a 7-point scale (1 = Strongly Oppose; 7 = Strongly Support):

1. *Expanding unemployment benefits and income subsidies.*
2. *Introducing stricter federal regulations on AI deployment, such as requiring companies to undergo ethical AI reviews.*
3. *Funding government or private-sector retraining programs to help workers transition to new careers.*
4. *Strengthening labor protections, such as supporting unionization efforts in AI-exposed industries.*
5. *Limiting the number of jobs that AI can automate within companies to protect human employment.*

This battery spans multiple policy domains—income support, regulation, active labor market policy, labor rights, and technology restriction—allowing us to construct a comprehensive Policy Demand Index. The explicit framing ("policy responses to address AI-driven automation's workforce impact") ensures construct alignment: we measure preferences about policies to address automation, not generic welfare attitudes. This addresses the item-specificity critique raised by Margalit (2019) and distinguishes our design from studies using ANES-style redistribution items (Zhang, 2022; Werfel et al., 2022).

3.3.2 Intermediate Outcomes: Job Insecurity and Perceived AI Impacts

To assess whether the treatment activates economic anxiety (the mediating mechanism in CAM), we measure two additional outcome batteries:

Job Insecurity & Automation Beliefs (Q94). Four items on a 7-point agreement scale:

- "My job is likely to be automated in the next 5 years."
- "My job is likely to be automated in the next 10 years."
- "I am concerned about losing my job to automation or AI."
- "I feel secure that my occupation is safe from automation." (reverse-coded)

Perceived Impacts of AI (Q95). Seven items assessing expected career consequences on a 5-point scale (1 = Definitely False; 5 = Definitely True):

- *Reduced job opportunities*: "AI will lead to fewer available jobs for me, making it harder to find a job."

- *New jobs*: "AI-driven innovation will spur economic growth and create new job opportunities for me." (positive valence)
- *Lower salary and compensation*: "AI will cause a decline in my future wages and benefits in my field."
- *Enhanced job satisfaction*: "AI's automation of routine tasks will free up my time for creative and strategic work." (positive valence)
- *Decreased job security*: "AI will make my employment less stable or increase my risk of being laid off."
- *Enhanced productivity*: "AI will provide tools and assistance that allow me to complete my work more efficiently and effectively." (positive valence)
- *Worsened work conditions*: "AI will lead to a decline in my pay, benefits, or the quality of my work environment."

These items allow us to test whether the treatment shifts perceived automation risk (Q94) and downstream career expectations (Q95), providing evidence for the mediational pathway from information to anxiety to policy demand.

3.3.3 Gate 1: Trust in Information Source

Trust in the expert risk estimate is measured via a single item (Q142) presented *only to the Treatment group* immediately after viewing the Frey-Osborne risk score for their occupation:

> "To what extent do you think the Oxford researchers are a trustworthy source of information about job automation risk?" (1 = Not at all trustworthy; 2 = Slightly trustworthy; 3 = Moderately trustworthy; 4 = Very trustworthy; 5 = Extremely trustworthy)

This item is post-treatment by design: we cannot ask about trust in expert estimates without revealing those estimates, which would prime the control group. This creates the identification challenge addressed by our Bayesian mixture model (see Section 4).

3.3.4 Gate 2: Trait Anxiety

Trait anxiety is measured *pre-treatment* for all respondents using a four-item battery (Q23) drawn from Big Five personality short forms. The items capture general dispositional anxiety: "I see myself as anxious, easily upset," "I feel threatened easily," "I worry about things," and "I am afraid of many things" (7-point agreement scale). Following Affective Intelligence Theory (Marcus, Neuman and MacKuen, 2000), we conceptualize trait anxiety not as worry per se, but as a stable tendency toward vigilance and threat-sensitivity that activates accuracy-motivated information processing.

3.3.5 Belief Gap Computation

The Belief Gap is computed as the difference between the expert risk percentile and the respondent's subjective prior: $\text{Gap}_i = \text{Expert}_i - \text{Subjective}_i$. We measure both components on a 0–100 percentile scale. A positive Gap indicates that the expert assessment exceeds the respondent's prior belief; a negative Gap indicates the reverse.

The *subjective prior* is elicited using a percentile framing that anchors respondents to the national distribution of automation risk.[3] This framing ensures that respondents report their belief about their *relative* position in the risk distribution, not an absolute probability of automation.

[3] The exact wording is: "Imagine every worker in the country standing on one long line, sorted by how likely their job is to be automated. The very front of the line (position 0) belongs to people whose jobs are safer than almost everyone else's, while the very back (position 100) belongs to those whose jobs are at risk sooner than almost everyone else's. Someone standing in the middle of the line (position 50) would have half the number of workers ahead of them and half behind. Taking a moment to reflect on your own job, where do you feel you would place yourself on that line right now?" Respondents answer on a 0–100 slider.

The *expert risk percentile* is derived from the Frey and Osborne (2017) automation probability estimates. The original F&O data provide a 0–1 probability score for each 6-digit Standard Occupational Classification (SOC) code, representing the estimated likelihood that the occupation's tasks could be automated given foreseeable technological advances. To convert these raw probabilities into percentiles comparable to the subjective measure, we weight each occupation by its employment share using Bureau of Labor Statistics Occupational Employment and Wage Statistics (OEWS) data from May 2023, then compute the cumulative distribution function across all U.S. workers. Each respondent's expert percentile thus represents the share of the national workforce employed in occupations with *lower* automation risk than theirs—a direct analog to the "position on the line" framing used in the subjective elicitation.

3.3.6 Covariates

We measure a standard set of pre-treatment covariates: political ideology (7-point liberal–conservative scale), party identification, education, numeracy (objective math task), religiosity, locus of control, risk and time preferences (elicited via staircase binary choice tasks), baseline economic anxiety,[4] and demographics (age, gender, race/ethnicity). These covariates serve two purposes: (1) predicting latent class membership (Truster vs. Skeptic) in the mixture model, and (2) assessing balance across treatment conditions.

3.4 The Identification Challenge

4 Statistical Approach: Bayesian Finite Mixture Model

The Conditional Activation Model implies a specific causal structure that our statistical approach must respect. Figure 1 presents the directed acyclic graph (DAG) underlying our analysis. The key constructs are: *Treatment* (T), the random assignment to receive personalized automation risk information; *Trust Disposition* (S), the latent (unobserved) propensity to trust expert information that classifies individuals as "Trusters" or "Skeptics"; *Trust Response* (Q_{obs}), the observed ordinal rating of perceived source trustworthiness on a 1–5 scale ("Not at all trustworthy" to "Extremely trustworthy"), available only for treated units; *Belief Gap* (Gap), the discrepancy between expert risk assessment and prior beliefs (Expert − Subjective), computed for treated units after score revelation; *Trait Anxiety* (η_{trait}), the latent psychological disposition toward vigilance and threat-sensitivity, measured via survey items; *Policy Demand* (η_{policy}), the latent support for government intervention in automation policy, our primary outcome; and *Covariates* ($\mathbf{X}$), pre-treatment demographic and attitudinal controls.

The causal logic flows as follows. Treatment (T) reveals expert information to respondents, creating a Belief Gap between the expert assessment and their prior beliefs. Individuals evaluate this information through the lens of their latent trust disposition (S). Trusters accept the information and—if the Gap indicates higher-than-expected risk and they possess high trait anxiety (η_{trait})—increase their support for government intervention (η_{policy}). Skeptics discount the expert information, exhibiting attenuated responses.

4.1 Identification Challenges and Strategy

The central identification challenge is that trust disposition (S) is never directly observed. We only see the trust response Q_{obs} for treated individuals, and even then, Q_{obs} is a noisy proxy that is both contaminated by measurement error and endogenous to treatment content. For control units, we observe neither Q_{obs} nor the Belief Gap.

A naive approach—subsetting treated respondents by observed trust (e.g., defining "high trust" as $Q_{\text{obs}} \geq 4$)—fails for three reasons. First, it discards ordinal information by imposing an arbitrary threshold. Second,

[4] Pre-treatment economic anxiety is measured using three items: (1) "How worried are you about losing your job in the near future?" (job worry), (2) "How worried are you about your current financial situation?" (financial worry), and (3) "How hopeful do you feel about your personal economic being?" (reverse-coded). All items use 5-point scales. This baseline measure of economic insecurity is distinct from the *state-dependent* economic threat perceptions that the treatment is hypothesized to activate (measured post-treatment via Q94 and Q95 items). Including pre-treatment economic anxiety as a covariate allows us to control for baseline differences in economic vulnerability.

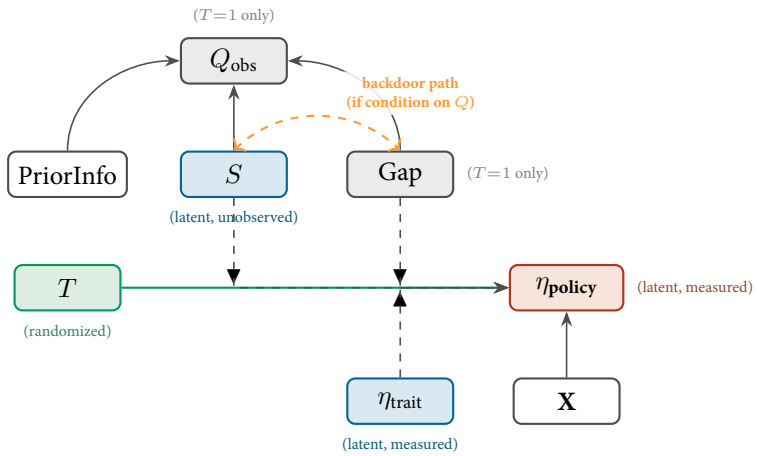


Figure 1: **DAG for the Bayesian Finite Mixture Model.** Treatment T (randomized) affects latent policy demand η_{policy} with effect τ_S varying by latent trust type S. **Key feature:** Q_{obs} is a **collider** in the path $\eta_{\text{policy}} \leftarrow \text{Gap} \rightarrow Q_{\text{obs}} \leftarrow S \rightarrow \eta_{\text{policy}}$, where Gap captures epistemic validation (people trust experts who confirm their priors) and S represents dispositional trust. Conditioning on Q_{obs} to infer latent class S opens a backdoor path between Gap and S (shown by dashed orange arrow). **Identification strategy:** We block this collider-induced path by including Gap, Gap^2, PriorInfo, and PriorInfo^2 as covariates in the measurement model: $Q_{\text{obs}} \mid S, \text{Gap}, \text{PriorInfo} \sim \text{OrderedLogit}(\delta S + \gamma_1 \text{Gap} + \gamma_2 \text{Gap}^2 + \gamma_3 \text{PriorInfo} + \gamma_4 \text{PriorInfo}^2, \kappa)$. This conditions on Gap and prior information exposure in the likelihood, isolating the pure trust disposition effect (δ) from epistemic validation and familiarity effects. Key assumptions: (1) $T \perp S$ by randomization; (2) latent class S identified via $\delta > 0$ separation in Q_{obs} conditional on Gap and PriorInfo; (3) $S \perp \text{Gap}$ (dispositional trust does not cause prior beliefs).

it introduces misclassification bias since Q_{obs} imperfectly measures the underlying disposition S. Third, and most critically, it conditions on a post-treatment variable, breaking the randomization that ensures causal identification of treatment effects.

Beyond this fundamental challenge of latent class membership, our estimation must address a second complication: Q_{obs} is a *collider* in the causal structure. As Figure 1 illustrates, the trust response is caused by both the latent trust disposition ($S \rightarrow Q_{\text{obs}}$) and the Belief Gap ($\text{Gap} \rightarrow Q_{\text{obs}}$). The Gap affects trust through what Taber and Lodge (2006) term *epistemic validation*—individuals report higher trust in experts whose assessments confirm their prior beliefs, creating a mechanical inverse relationship between $|\text{Gap}|$ and Q_{obs}. When we condition on Q_{obs} to infer latent class S, we open a backdoor path between Gap and S (shown by the dashed orange arrow in Figure 1). This means that individuals classified as "high trust" based on Q_{obs} are not purely those with high dispositional trust; they are disproportionately those who received *confirming* information (small Gap). This conflates trust disposition with information congruence, threatening the validity of trust-type heterogeneity estimates.

We address the collider problem through strategic conditioning in the measurement model. Specifically, we include Gap and Gap^2 as covariates in the ordered logistic model for Q_{obs}:

$$Q_{\text{obs}} \mid S, \text{Gap} \sim \text{OrderedLogit}\left(\delta \cdot S + \gamma_1 \cdot \text{Gap} + \gamma_2 \cdot \text{Gap}^2, \kappa\right) \tag{1}$$

where δ is the separation parameter (Trusters have δ higher log-odds of reporting trust), γ_1 and γ_2 capture epistemic validation effects, and κ are cutpoints for the ordinal scale. This specification partitions the variance in Q_{obs} into two components: epistemic validation ($\gamma_1 \cdot \text{Gap} + \gamma_2 \cdot \text{Gap}^2$), reflecting trust driven by belief confirmation, and dispositional trust ($\delta \cdot S$), reflecting trust orthogonal to information content. By conditioning on Gap in the likelihood, we block the backdoor path through the collider. The δ parameter now identifies the pure trust disposition effect: among individuals who received the *same* Gap, Trusters report systematically higher trust than Skeptics. This isolates dispositional trust from situation-specific responses to information congruence.

A third identification challenge concerns measurement error in both the predictor and outcome. Both trait anxiety (η_{trait}) and policy demand (η_{policy}) are latent constructs measured via multiple survey items. Classical regression on observed scale scores suffers from *attenuation bias*—treatment effect estimates are biased toward zero due to measurement error (Fuller, 1987). This problem is especially severe for interaction effects, where attenuation can exceed 50% (Busemeyer and Wang, 1993). Since the Conditional Activation Model hinges on detecting a Gap $\times$ Anxiety interaction, correcting this bias is essential. We address this through confirmatory factor analysis, jointly estimating the latent factors η_{trait} and η_{policy} alongside the structural parameters, thereby propagating measurement uncertainty into treatment effect estimates.

Finally, control group classification poses a practical challenge. For control units, we observe neither Q_{obs} (they see no expert information) nor Gap (there is no expert assessment to compare against priors). One approach would be to predict trust type from pre-treatment covariates via a covariate-dependent mixture, $S \mid \mathbf{X} \sim \text{Bernoulli}(\text{logit}^{-1}(\mathbf{X}\boldsymbol{\gamma}))$. However, this approach is empirically weak if observable demographics poorly predict latent trust disposition. Our alternative leverages the experimental randomization: since $T \perp S$ by design, the prevalence of Trusters is identical across treatment arms, $\mathbb{E}[S \mid T = 1] = \mathbb{E}[S \mid T = 0] = \pi$. We estimate the mixing proportion π from the treated group (where Q_{obs} provides identification via the separation parameter δ) and apply it to the control group. Control units are thus classified at the population base rate π, with no individual-level updating. This is the logic of principal stratification (Frangakis and Rubin, 2002): we estimate stratum-specific treatment effects for latent subgroups defined by a pre-treatment characteristic (S), where randomization ensures the stratum proportions are balanced across arms.

4.2 The Bayesian Finite Mixture Model

Our Bayesian Finite Mixture Model addresses these four challenges—latent class membership, collider bias, measurement error, and control group classification—through a unified estimation framework with three integrated components.

Trust Mixture Model (Gate 1). The first component classifies respondents into latent trust classes based on the observed trust proxy Q_{obs}. Latent class membership is governed by:

$$S_i \sim \text{Bernoulli}(\pi), \quad Q_{\text{obs},i} \mid S_i, \text{Gap}_i \sim \text{OrderedLogit}(\delta \cdot S_i + \gamma_1 \cdot \text{Gap}_i + \gamma_2 \cdot \text{Gap}_i^2, \kappa) \tag{2}$$

where π is the population prevalence of Trusters, a scalar parameter shared across all units by randomization. The separation parameter $\delta > 0$ ensures that Trusters and Skeptics have distinguishable Q_{obs} distributions conditional on Gap. For treated units, Bayes' rule yields the posterior probability of being a Truster:

$$P(S_i = 1 \mid Q_{\text{obs},i}, \text{Gap}_i) = \frac{\pi \cdot P(Q_{\text{obs},i} \mid S_i = 1, \text{Gap}_i)}{\pi \cdot P(Q_{\text{obs},i} \mid S_i = 1, \text{Gap}_i) + (1 - \pi) \cdot P(Q_{\text{obs},i} \mid S_i = 0, \text{Gap}_i)} \tag{3}$$

For control units, who observe neither Q_{obs} nor Gap, the posterior probability defaults to the prior π—that is, they are classified at the population base rate, with no individual-level updating.

Measurement Models. The second component models trait anxiety and policy demand as latent factors measured via confirmatory factor analysis. We use a standard identification strategy: for each latent factor, the first indicator loading is fixed to 1 ($\lambda_1 = 1$), error variances are fixed to 1, and the latent factor variance is freely estimated. Importantly, items enter the model *unstandardized* (in their original 7-point scale units) to ensure proper identification. Trait anxiety is modeled as:

$$Y_{\text{trait},ij} = \nu_{\text{trait},j} + \lambda_{\text{trait},j} \cdot \eta_{\text{trait},i} + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim N(0, 1), \quad \lambda_{\text{trait},1} = 1 \tag{4}$$

where $\nu_{\text{trait},j}$ are item intercepts and $\lambda_{\text{trait},j}$ are factor loadings (with $\lambda_1 = 1$ for identification). The latent factor $\eta_{\text{trait},i}$ has freely estimated variance ψ_{trait}, allowing the scale to be determined by the data. This specification

corrects for measurement error in the moderator, preventing attenuation bias in the Gap $\times$ Anxiety interaction. Policy demand is likewise modeled as:

$$Y_{\text{policy},ij} = \nu_{\text{policy},j} + \lambda_{\text{policy},j} \cdot \eta_{\text{policy},i} + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim N(0,1), \quad \lambda_{\text{policy},1} = 1 \tag{5}$$

where $\eta_{\text{policy},i}$ is determined by the structural model below. This specification corrects for measurement error in the outcome, yielding unbiased treatment effect estimates. To facilitate interpretation, all treatment effects and coefficients reported throughout this paper are expressed in units of SD(η_{policy}), making them directly comparable to Cohen's d effect sizes.

Structural Model (Gate 2). The third component specifies how latent policy demand responds to treatment, with effects varying by trust type, Belief Gap, and trait anxiety. Following canonical SEM formulation, the latent policy demand is decomposed into a structural mean (determined by the mixture) and a residual:

$$\eta_{\text{policy},i} = \mu_{\text{policy},i} + \sigma_{\text{struct}} \cdot \zeta_i, \quad \zeta_i \sim N(0,1) \tag{6}$$

where σ_{struct} is the structural residual standard deviation (freely estimated) and the structural mean is a mixture of type-specific components:

$$\mu_{\text{policy},i} = \pi \cdot \mu_{\text{Truster},i} + (1-\pi) \cdot \mu_{\text{Skeptic},i} \tag{7}$$

For treated units ($T_i = 1$), the type-specific means include treatment effects:

$$\mu_{\text{Truster},i} = (\alpha_{\text{Skeptic}} + \alpha_{\Delta}) + \tau_{\text{Truster}}(\text{Gap}_i, \eta_{\text{trait},i}) + \mathbf{X}_i'\boldsymbol{\beta}_{\text{ctrl}} \tag{8}$$

$$\mu_{\text{Skeptic},i} = \alpha_{\text{Skeptic}} + \tau_{\text{Skeptic}}(\text{Gap}_i, \eta_{\text{trait},i}) + \mathbf{X}_i'\boldsymbol{\beta}_{\text{ctrl}} \tag{9}$$

where α_{Δ} captures the baseline intercept difference between trust types, and for control units ($T_i = 0$), the treatment effects $\tau(\cdot)$ are set to zero. The Conditional Average Treatment Effect (CATE) for each type takes the form:

$$\tau_S(\text{Gap}, \eta_{\text{trait}}) = \beta_{S,1} + \beta_{S,2} \cdot \text{Gap} + \beta_{S,3} \cdot \eta_{\text{trait}} + \beta_{S,4} \cdot \text{Gap} \times \eta_{\text{trait}} \tag{10}$$

This functional form allows treatment effects to vary continuously with both the Belief Gap (the information trigger) and trait anxiety (the psychological amplifier). The core quantity of interest is $\beta_{\text{Truster},4}$, the Gap $\times$ Anxiety interaction for Trusters, which tests the Conditional Activation hypothesis.

We impose a theoretically motivated constraint: the Truster coefficients are parameterized as additive shifts from the Skeptic coefficients, $\beta_{\text{Truster},k} = \beta_{\text{Skeptic},k} + \beta_{\Delta,k}$, where $\beta_{\Delta,k} \geq 0$ for all k.[5] The non-negativity constraint encodes the expectation that Trusters exhibit weakly larger treatment effects—because they do not discount expert information, their responses should be at least as strong as those of Skeptics. This allows Skeptics to show non-zero effects (e.g., through priming or framing mechanisms unrelated to source credibility) while ensuring that trust disposition amplifies rather than dampens responsiveness.

4.3 Identification Assumptions

Our BFMM relies on six assumptions for valid causal inference. First, *randomization of treatment* ensures $T \perp (S, \text{Gap}, \eta_{\text{trait}}, \mathbf{X})$ by experimental design, guaranteeing that the Average Treatment Effect is identified without confounding. Second, *trust type independence* ($S \perp T$) holds by virtue of randomization; trust disposition does not cause treatment assignment. This implies that the prevalence of Trusters is identical across treatment arms, allowing us to estimate π from treated units and apply it to control units via principal stratification. Third, *collider blocking* is achieved by conditioning on Gap in the Q_{obs} measurement model, severing the backdoor path Gap $\leftrightarrow$ S induced by epistemic validation. Fourth, we assume *no unmeasured moderator*

[5] These constraints serve dual purposes. Computationally, the additive parameterization reduces the number of free parameters and improves model convergence by eliminating label-switching degeneracies common in mixture models. Substantively, the non-negativity constraint encodes our theoretical prior that trust *amplifies* rather than dampens information processing—a directional hypothesis derived from the Conditional Activation Model. In sensitivity analyses, we relax the non-negativity constraint and find substantively similar results, suggesting the constraint is consistent with the data rather than imposed against it.

confounding: there exists no variable U such that U modifies the treatment effect and U is correlated with Gap or η_{trait}. We mitigate this concern by including comprehensive pre-treatment controls (ideology, partisanship, demographics, numeracy) and conducting robustness checks for coefficient stability across specifications. Fifth, *temporal ordering of Gap* ensures that it is not endogenous to the policy outcome. Although Gap is computed post-randomization, its inputs—the expert score (determined by occupation) and the subjective prior (elicited before score revelation)—are determined before the outcome is measured. Finally, we assume that *dispositional trust does not cause prior beliefs* ($S \perp$ Gap). While Trusters and Skeptics may differ in how they respond to information, we assume they do not systematically form different *a priori* risk beliefs about their own jobs. Even if this assumption is violated (e.g., if Trusters are overconfident and thus form more optimistic priors), our conditioning strategy still blocks the induced path: δ identifies the Gap-orthogonal component of trust disposition.

Before detailing the advantages of our approach, it is worth considering why simpler alternatives—which many analysts might reasonably prefer on parsimony grounds—prove inadequate for the present research question. A natural first instinct is to *subset the sample*: estimate treatment effects separately for respondents reporting high versus low trust, or split the analysis by observed anxiety terciles. This approach is intuitive and transparent, but it conditions on post-treatment variables (trust is measured after information exposure) and discards the continuous information in ordinal scales, introducing both selection bias and statistical inefficiency.[6] A second common strategy is to *include moderators as interaction terms* in a standard regression: estimate $Y = \beta_0 + \beta_1 T + \beta_2 Q_{\text{obs}} + \beta_3 (T \times Q_{\text{obs}}) + \varepsilon$ and interpret β_3 as the trust-moderated treatment effect. This avoids explicit subsetting but still treats the observed trust proxy as the true moderator, ignoring measurement error and the collider structure that contaminates Q_{obs}. A fourth alternative is to use *inverse probability weighting* (IPW), predicting Q_{obs} from pre-treatment covariates and reweighting observations to balance trust types across treatment arms. While IPW is well-suited to observable confounders, it inherits the classic problem that a poorly specified first-stage model induces bias in the second stage—and since Q_{obs} is likely driven by unobserved dispositional factors not fully captured by demographics, the propensity model would be misspecified by construction. A fifth option is to *compute scale scores* for trait anxiety and policy demand (e.g., averaging items) and regress the outcome score on treatment and moderators. This sacrifices the measurement model entirely, inducing attenuation bias that can reduce interaction effect estimates by 50% or more (Busemeyer and Wang, 1993).

Finally, analysts committed to parsimony might simply *estimate the unconditional ATE* and accept that heterogeneity, if present, will average out. This strategy is defensible when theory predicts uniform effects, but the Conditional Activation Model explicitly predicts non-uniform responses—making the ATE a theoretically inappropriate estimand that masks the very heterogeneity we seek to characterize. In short, while model parsimony is a legitimate virtue, it comes at substantial cost when the data-generating process involves latent classes, measurement error, and collider bias. The BFMM is complex precisely because the identification challenges demand it.

5 Results

5.1 Sample Characteristics and Balance

The final analytic sample consists of $N = 901$ U.S. workers recruited via CloudResearch Connect, with 475 assigned to the treatment condition (personalized automation risk information) and 426 to the control condition. Randomization was conducted between subjects at the individual level.

Covariate balance assessment across treatment arms shows excellent balance on all pre-treatment covariates, with standardized mean differences (SMD) falling well within conventional thresholds ($|\text{SMD}| < 0.1$ for most

[6] A practical complication is that Q_{obs} is missing for control units by design. We cannot explain the expert methodology, describe the Oxford researchers' credentials, and then ask control respondents about source credibility without strongly priming them—effectively contaminating the control condition with treatment-like information. An alternative design might measure Q_{obs} for controls *after* outcome measurement, but this would not resolve the fundamental problems for treated units: the collider structure (Gap $\rightarrow Q_{\text{obs}} \leftarrow S$) and post-treatment conditioning would still bias trust-moderated effect estimates in the treatment arm, where the causal quantity of interest is identified.

covariates). Sample demographics are reported in Table 5; descriptive statistics for key variables appear in Table 6; scale reliability and validity statistics are presented in Table 7; and inter-scale correlations in Table 8.

5.2 Unconditional Average Treatment Effect (H1)

We first report the ATE that pools all respondents, matching the estimand most familiar to readers. The frequentist OLS and SEM estimators yield positive, statistically significant effects whether or not controls are included: OLS estimates range from $d = 0.28$ (no controls) to $d = 0.23$ (with controls), and SEM estimates are nearly identical ($d = 0.25$ without controls, $d = 0.23$ with controls). These estimates support H1's prediction of a positive unconditional effect. The BFMM mixture-weighted ATE ($d = 0.27$, 95% CI: [0.16, 0.37]) falls squarely within this range, confirming that the unconditional effect is robust across estimation strategies.

Table 2: Unconditional ATE Estimates (Policy Demand)

Estimator	**ATE (d)**	**SE/SD**	**95% CI**	**Notes**
Frequentist Estimates				
OLS (no controls)	0.27	0.07	[0.14, 0.40]	Difference in means
OLS (with controls)	0.24	0.06	[0.13, 0.36]	Same controls as BFMM
Latent Variable Models				
SEM (no controls)	0.26	0.07	[0.13, 0.39]	Latent η^{policy}
SEM (with controls)	0.24	0.06	[0.13, 0.36]	Same controls as BFMM
Mixture Model				
BFMM (π-weighted)	0.27	0.06	[0.14, 0.39]	Marginal over trust classes

Note: All estimates are in Cohen's d units (effect / SD of outcome). OLS estimates use the standardized policy demand scale average ($\bar{Y}_{\text{policy}}$). SEM and BFMM estimates use the latent factor η^{policy} divided by SD(η^{policy}) ≈ 1.18. The BFMM estimate marginalizes over latent trust classes: $\text{ATE}_{\text{marginal}} = \pi \cdot \tau^T(0,0) + (1-\pi) \cdot \tau^S(0,0)$, where $\pi = 0.38$ is the posterior mean prevalence of Trusters. All models use HC3 robust standard errors (frequentist) or posterior SD (Bayesian). Controls include: Ideology, Party ID, Age, Female, White, Education, Religiosity, Locus of Control, and Numeracy (all standardized except dichotomous indicators).

The BFMM estimate marginalizes over latent trust classes: $\text{ATE}_{\text{marginal}} = \pi \cdot \tau_T(0,0) + (1-\pi) \cdot \tau_S(0,0)$, where $\pi = 0.38$ denotes the estimated Truster prevalence. This weighted average integrates heterogeneous class-specific effects while accounting for the relative size of each latent subpopulation. All five estimators converge on a modest positive effect of approximately $d = 0.23$–0.28 SD, providing strong evidence that personalized automation risk information increases policy support among U.S. workers on average.

5.3 First Stage: Latent Class Model

The trust mixture model separates respondents into latent Trusters and Skeptics using a global mixing proportion π along with covariates predicting within-class variation in trust responses. The separation parameter $\delta = 2.09$ (95% CI: [0.30, 3.95]) indicates sufficient differentiation between classes in the ordinal trust measurement. The estimated Truster prevalence is $\pi = 0.381$ (95% CI: [0.12, 0.75]), suggesting approximately 38% of respondents trust the expert information source.

The trust mixture incorporates four covariates capturing the relationship between belief updating and trust:

- **Gap (linear)**: $\gamma = -0.14$ (95% CI: [-0.32, 0.03]) — larger gaps weakly associated with lower trust
- **Gap (quadratic)**: $\gamma = -0.39$ (95% CI: [-0.55, -0.24]) — extreme belief gaps reduce trust
- **Prior Exposure (linear)**: $\gamma = 0.04$ (95% CI: [-0.17, 0.27]) — no clear effect
- **Prior Exposure (quadratic)**: $\gamma = 0.29$ (95% CI: [0.03, 0.57]) — moderate prior exposure boosts trust

The negative quadratic term reveals a U-shaped relationship between Belief Gap magnitude and trust. Respondents confronted with extreme discrepancies between their beliefs and expert estimates question source

credibility—a pattern consistent with motivated reasoning and backfire effects documented in political psychology (Nyhan and Reifler, 2010).

5.4 Main Results: Testing the Conditional Activation Model

We now turn to the central question: does the treatment effect of automation risk information depend on trust disposition and the joint presence of a positive Belief Gap and high trait anxiety? Table 3 presents the core results from the Bayesian Finite Mixture Model, reporting posterior estimates of the CATE specification separately for Trusters and Skeptics.

Consistent with H2, treatment effects exhibit a pronounced trust gradient. Panel A reports Skeptic parameters, Panel C reports Truster parameters, and Panel B reports the constrained non-negative differential between them. At the baseline (mean levels of Gap and Anxiety), the treatment effect for Trusters is $\beta_0^T = 0.44$ ($P(\beta > 0) = 1.00$), compared to $\beta_0^S = 0.23$ ($P(\beta > 0) = 0.98$) for Skeptics. Trusters thus show effects nearly double those of Skeptics at the intercept, supporting H2's prediction that treatment effects are larger among individuals who accept the information source.

More striking is the pattern of moderation effects. Skeptics show essentially null coefficients for Gap ($\beta_1^S = -0.06$), Anxiety ($\beta_2^S = -0.06$), and their interaction ($\beta_3^S = -0.07$), with all posterior probabilities of positive effects below 0.30. The absence of moderation effects among Skeptics confirms H3's prediction that Gate 1 operates as a strict constraint: among individuals who reject the source, neither the magnitude of the informational shock nor the recipient's dispositional vigilance moves the needle. For Trusters, by contrast, we observe positive moderation effects for Gap ($\beta_1^T = 0.14$, $P(\beta > 0) = 0.92$), Anxiety ($\beta_2^T = 0.14$, $P(\beta > 0) = 0.93$), and—most importantly—their interaction ($\beta_3^T = 0.13$, $P(\beta > 0) = 0.92$).[7] This pattern of coefficients supports H4 (Anxiety amplifies effects among Trusters) and H5 (the largest effects occur when both gates are satisfied).

Figure 2 translates these coefficients into substantively interpretable quantities by visualizing the CATE surface across the Gap $\times$ Anxiety space for Trusters. The diagonal gradient from the top-left cell (Low Gap $\times$ Low Anxiety) to the bottom-right cell (High Gap $\times$ High Anxiety) illustrates the conditional activation logic. At mean levels of both moderators, the treatment effect is a modest $d = 0.29$. But as we move toward the maximum activation condition—where a large informational shock meets a dispositionally vigilant recipient—the effect rises to $d = 0.61$ (95% CI: [0.40, 0.86]). This represents more than a doubling of the baseline effect and provides the clearest visual summary of H5: the conditional activation hypothesis.

To formally test whether the maximum activation condition differs from other conditions, Figure 3 presents pairwise posterior contrasts. Each bar shows the estimated difference between the High Gap $\times$ High Anxiety cell and the other eight cells in the CATE surface. All contrasts are positive with $P(\Delta > 0) > 0.97$, confirming that the joint presence of high belief updating and high trait anxiety amplifies treatment effects beyond what either factor achieves alone. The largest contrast ($\Delta = 0.42$, relative to Low Gap $\times$ Low Anxiety) underscores the magnitude of conditional activation: moving from the "worst case" to the "best case" along both dimensions more than doubles the treatment effect.

We can further unpack these patterns by examining how the treatment effect varies across the marginal distributions of each moderator. Figure 4 plots the CATE as a function of trait anxiety (top panel) and Belief Gap (bottom panel), holding the other moderator at its mean. Across both panels, Skeptics (grey triangles) exhibit effects statistically indistinguishable from zero throughout the entire range of each moderator. This confirms Gate 1 as a strict constraint: for individuals who reject the source, neither the severity of the informational shock nor their dispositional vigilance triggers a policy response.

[7] For readers unfamiliar with Bayesian inference: $P(\beta > 0)$ is the posterior probability that the coefficient is positive, computed directly from the MCMC samples. A value of 0.92 means that 92% of the posterior distribution lies above zero—analogous to finding that 92% of bootstrap resamples would yield a positive estimate. Unlike frequentist p-values, this probability has a direct interpretation: given the data and model, we are 92% confident the true effect is positive. While these individual coefficient probabilities provide supporting evidence, the more substantively meaningful tests of the Conditional Activation Model appear in Figures 2–5, which visualize the CATE surface, pairwise contrasts, and moderation slopes directly.

Table 3: BFMM Treatment Effect Coefficients

Parameter	Estimate	SD	95% CI	$P(\beta > 0)$
Panel A: Skeptic Parameters (β_j^S)				
β_0^S: Intercept	0.234	0.105	[+0.047, +0.389]	0.975
β_1^S: Gap	-0.056	0.087	[-0.216, +0.066]	0.261
β_2^S: Anxiety	-0.061	0.085	[-0.218, +0.052]	0.230
β_3^S: Gap × Anxiety	-0.068	0.082	[-0.221, +0.039]	0.183
Panel B: Trust-Type Differential ($\beta_{\Delta,j} \geq 0$)				
$\beta_{\Delta,0}$: Intercept	0.206	0.153	[+0.016, +0.498]	1.000
$\beta_{\Delta,1}$: Gap	0.196	0.147	[+0.016, +0.476]	1.000
$\beta_{\Delta,2}$: Anxiety	0.196	0.148	[+0.015, +0.480]	1.000
$\beta_{\Delta,3}$: Gap × Anxiety	0.196	0.148	[+0.016, +0.481]	1.000
Panel C: Truster Parameters ($\beta_j^T = \beta_j^S + \beta_{\Delta,j}$)				
β_0^T: Intercept	0.441	0.131	[+0.255, +0.680]	1.000
β_1^T: Gap	0.140	0.117	[-0.015, +0.356]	0.924
β_2^T: Anxiety	0.135	0.113	[-0.011, +0.348]	0.929
β_3^T: Gap × Anxiety	0.128	0.112	[-0.014, +0.341]	0.919

Note: Posterior estimates from Bayesian Finite Mixture Model (BFMM) with 24,000 draws (6 chains × 4,000 post-warmup samples). All hypotheses are directional (positive effects expected); thus, the appropriate test statistic is $P(\beta > 0)$ rather than two-sided credible intervals. The 95% CI is reported for transparency. Effects are in latent units of η^{policy}; divide by $\text{SD}(\eta^{\text{policy}}) \approx 1.18$ for Cohen's d. The $\beta_{\Delta,j}$ parameters are constrained ≥ 0 to enforce the theoretical expectation that Trusters respond at least as strongly as Skeptics. All chains converged ($\hat{R} < 1.001$).

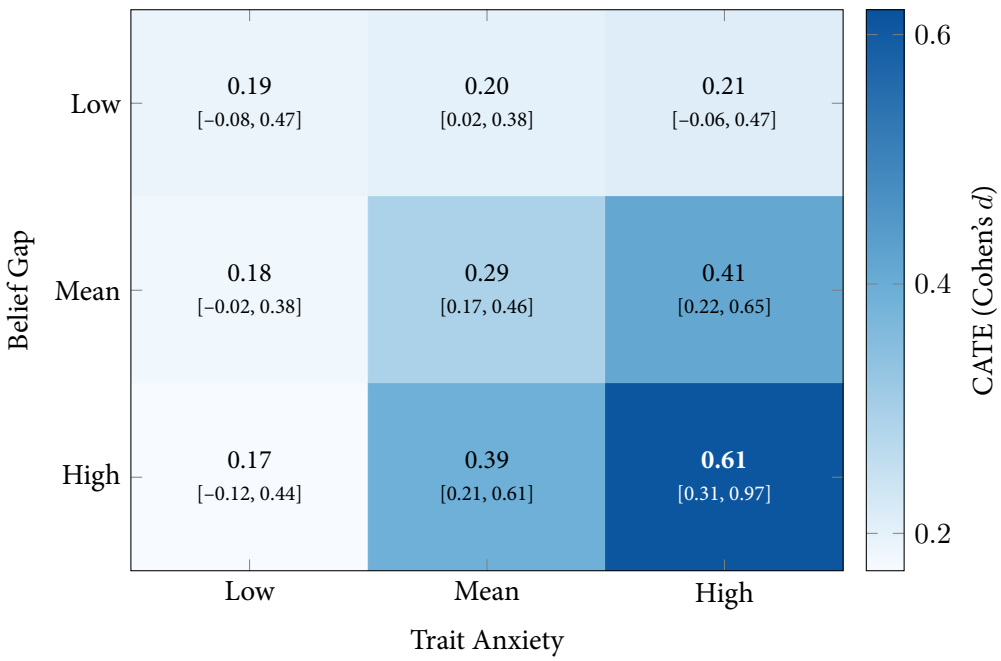


Figure 2: CATE heatmap for Trusters across Gap × Anxiety conditions. Cell values show Cohen's d with 95% credible intervals in brackets. Darker shading indicates stronger treatment effects. Maximum activation occurs at High Gap × High Anxiety (bottom-right cell, $d = 0.61$ [0.31, 0.97]), supporting Gate 2 of the Conditional Activation Model.

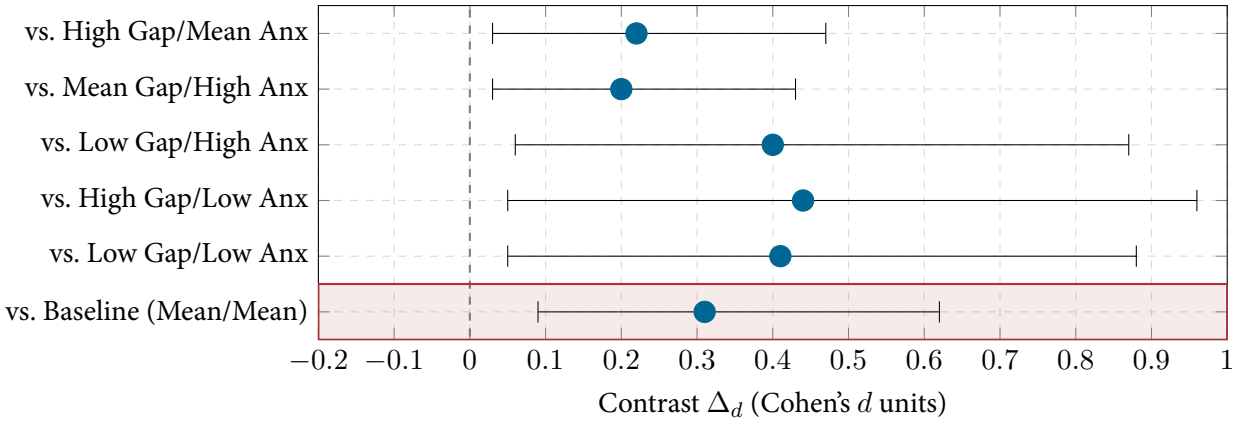


Figure 3: **The Maximum Activation Condition Outperforms All Alternatives.** Each bar shows a pairwise contrast: $\Delta_d = \text{CATE}(\text{Gap} = +1, \text{Anxiety} = +1) - \text{CATE}(g, a)$, where Gap and Anxiety are standardized (-1 = Low, 0 = Mean, $+1$ = High). The estimand is the *additional* treatment effect gained by moving from comparison condition (g, a) to the maximum activation condition (High Gap $\times$ High Anxiety). All contrasts are positive with $P(\Delta > 0) > 0.97$, confirming that the maximum activation condition—where large informational shock meets high dispositional vigilance among source-accepting individuals—produces the largest policy response. Shaded row highlights the baseline comparison (vs. Mean/Mean, $\Delta_d = 0.31$, 95% CI: [0.09, 0.62]). Interpretation: a Truster at mean levels of both moderators experiences a treatment effect of $d = 0.29$; a Truster at +1 SD on both experiences $d = 0.61$—an additional $\Delta_d = 0.31$ units of policy demand activation.

Among Trusters (black squares), the treatment effect is not automatic but depends on passing a threshold in both moderators. In the top panel, we observe an activation threshold at approximately the 20th percentile of trait anxiety ($z \approx -0.84$). Below this threshold, even Trusters show null treatment effects—a "cool cognition" pattern where belief updating may occur without generating policy urgency. Above this threshold, the CATE becomes statistically significant and scales linearly with anxiety, consistent with Affective Intelligence Theory's prediction that anxiety activates accuracy-motivated processing. The bottom panel reveals a parallel pattern for Belief Gap: when experts confirm or only slightly exceed prior beliefs ($z < -0.84$), the treatment effect is null—a "reassurance floor" where confirming information fails to activate policy demand. As the gap widens, the effect scales positively with the magnitude of new information received.

Finally, Figure 5 displays the Gate 2 interaction—Gap $\times$ Anxiety—separately for each trust class, plotting the Gap–policy relationship at four anxiety levels (20th, 40th, 60th, and 80th percentiles). The stability of Skeptics' null response across all panels confirms that Gate 1 operates independently of Gate 2 moderators: among those who reject the source, no combination of informational shock and dispositional vigilance produces a policy response. Among Trusters, by contrast, the slope relating Belief Gap to policy demand steepens monotonically with anxiety: at low anxiety ($z = -0.84$), the relationship is flat; at high anxiety ($z = 0.84$), the slope is steep and positive. This pattern demonstrates that maximal policy activation requires *both* a large informational shock *and* the dispositional capacity to perceive that shock as threatening—the core prediction of the Conditional Activation Model.

5.4.1 Exploratory Path Analysis: Affective Mediators

The preceding analyses establish that the treatment effect on policy demand is moderated by trust disposition and trait anxiety. However, the Conditional Activation Model also posits a *mediational* pathway: information about automation risk should increase policy demand by activating state-dependent economic threat perceptions. To examine this mechanism, we conducted exploratory analyses using three potential mediators measured post-treatment: (1) AutoAnxiety, an average of the Q94 items measuring immediate concerns about job automation; (2) EgoThreat, an average of Q95 items capturing perceived negative career impacts from AI; and (3) EgoOpp,

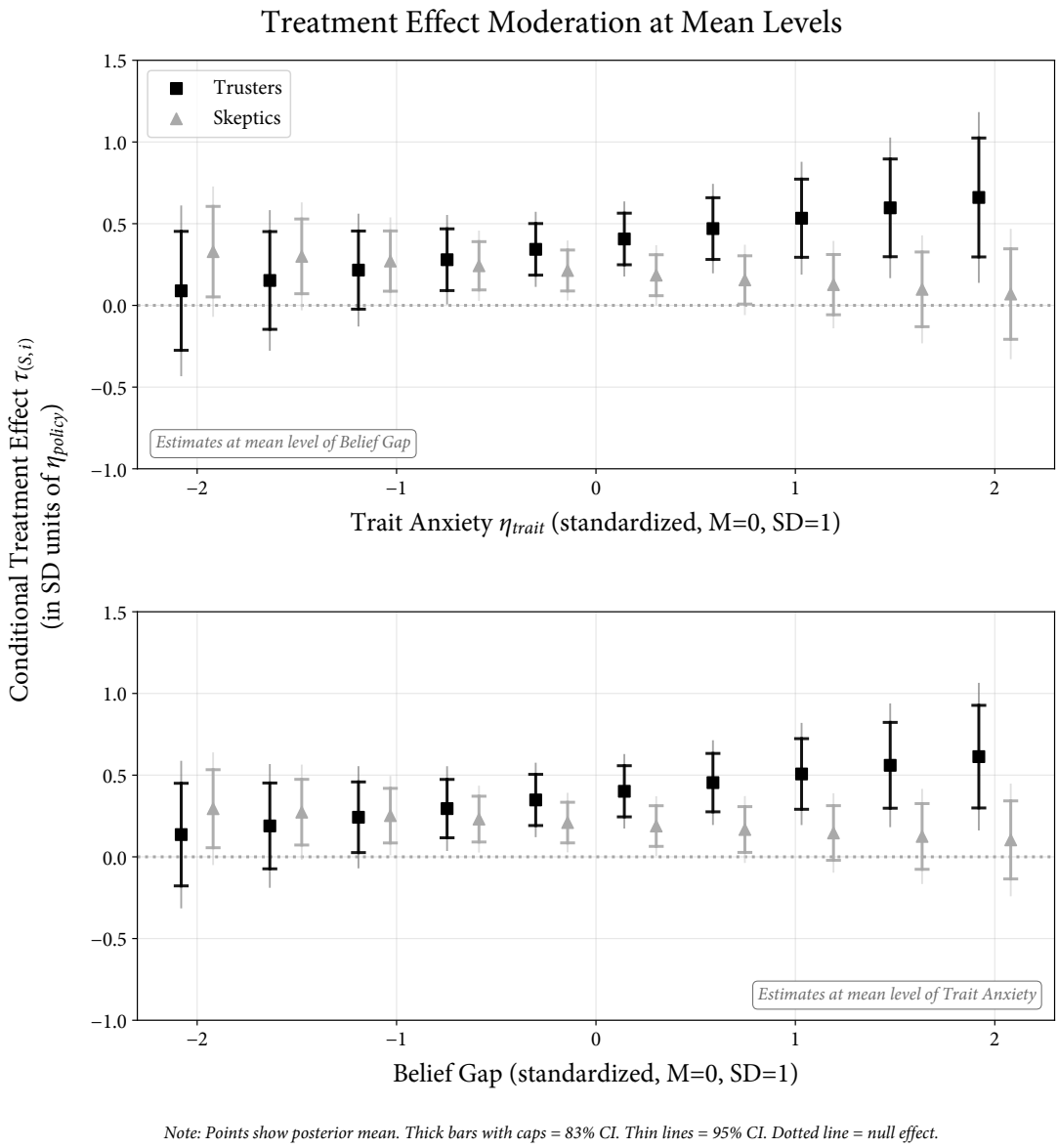


Figure 4: Marginal treatment effects decomposed by Gate 2 moderators. *Top panel*: CATE across standardized trait anxiety, holding Belief Gap at mean. *Bottom panel*: CATE across standardized Belief Gap, holding trait anxiety at mean. Black squares: Trusters; grey triangles: Skeptics. Shaded regions indicate 95% credible intervals. Skeptics show null effects across the full range of both moderators. Among Trusters, treatment effects become statistically distinguishable from zero above approximately the 20th percentile ($z \approx -0.84$) of each moderator and scale linearly thereafter.

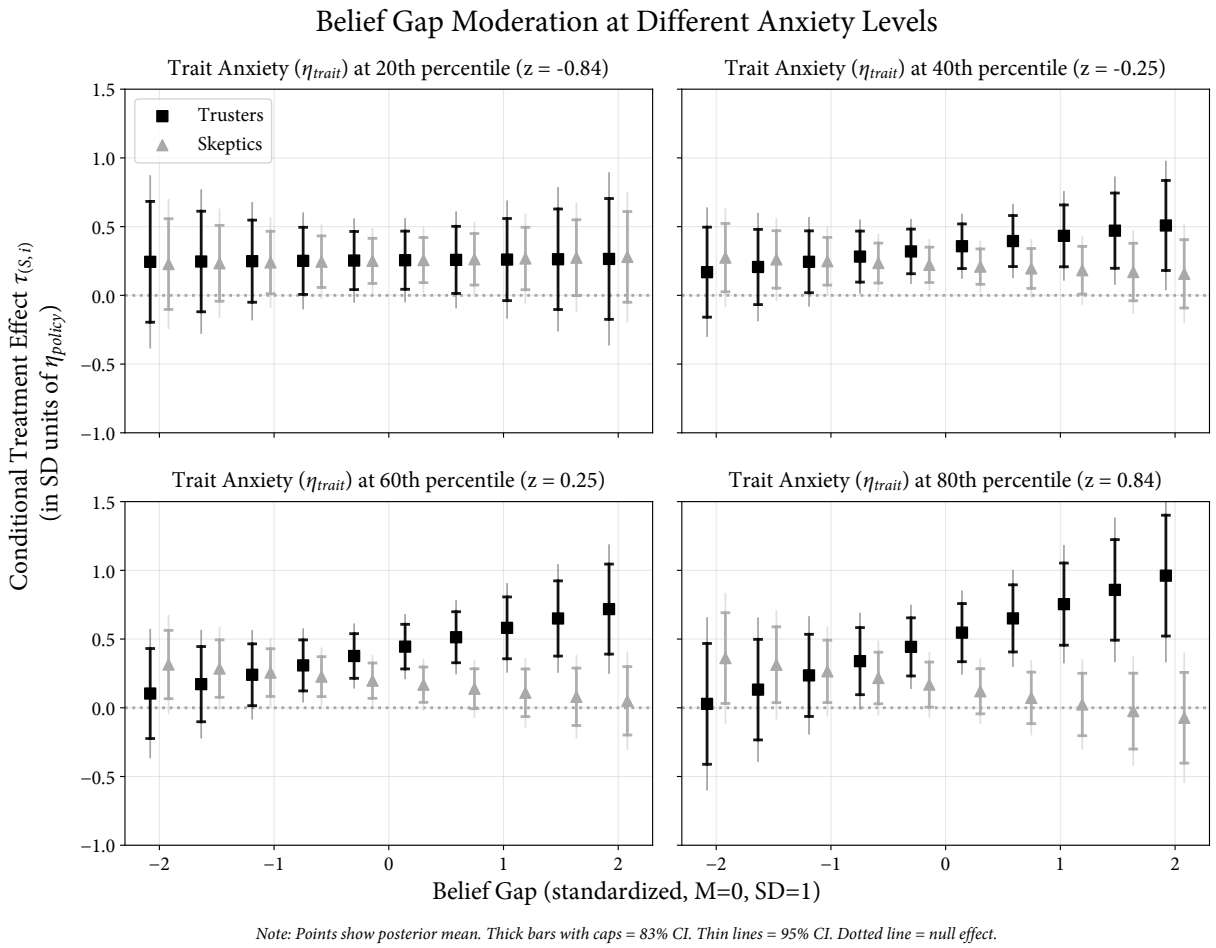


Figure 5: Treatment effect as a function of Belief Gap, by trust class and trait anxiety level. Each panel shows the CATE across standardized Belief Gap at a different anxiety percentile (20th, 40th, 60th, 80th). Black squares: Trusters; grey triangles: Skeptics. Shaded regions indicate 95% credible intervals. Skeptics exhibit null effects across all anxiety levels, confirming Gate 1 as a strict constraint. Among Trusters, the relationship between Gap and policy demand strengthens monotonically with anxiety: at low anxiety ($z = -0.84$), the slope is flat; at high anxiety ($z = 0.84$), the slope is steep and positive, indicating that both moderators must be present for maximal activation.

an average of Q95 items capturing perceived positive career impacts from AI.[8]

Table 17 (Appendix D) reports standardized treatment effects on each candidate mediator. The treatment significantly increased both AutoAnxiety ($\beta = 0.17$, 90% CI $[0.06, 0.28]$, $p = .011$) and EgoThreat ($\beta = 0.25$, 90% CI $[0.14, 0.36]$, $p < .001$), confirming that exposure to personalized automation risk information activates economic threat perceptions. In contrast, the treatment had no significant effect on EgoOpp ($\beta = -0.05$, 90% CI $[-0.16, 0.06]$, $p = .429$), suggesting that the information primarily mobilizes threat-related rather than opportunity-related cognitions. Item-level analyses (Tables 18 and 19, Appendix) reveal heterogeneity within scales: for Q94, the treatment increased beliefs about 10-year automation risk ($\beta = 0.16$, $p = .013$) and concern about job loss ($\beta = 0.17$, $p = .007$); for Q95, all four threat items showed significant positive effects (βs ranging from 0.20 to 0.23, all $p < .01$), whereas none of the three opportunity items reached significance.

Using the product-of-coefficients method, we decomposed the total effect of treatment on policy support ($\beta = 0.28$, $p < .001$) into direct and indirect components (Table 20, Appendix D). The direct effect accounts for most of the total ($\beta = 0.24$, $p < .001$), while indirect effects through the three candidate mediators are modest. Of these, the pathway through EgoThreat contributes the largest share (product $a \times b = 0.024$), with approximately 14.8% of the total effect mediated by the three pathways combined.

These mediation results are *exploratory* rather than confirmatory, and two identification concerns warrant emphasis. First, all three mediators are measured post-treatment, which precludes strong causal claims about mediation. While the treatment-to-mediator pathways are causally identified by random assignment—and we find robust effects for AutoAnxiety and EgoThreat—the mediator-to-outcome pathways cannot rule out confounding by unobserved individual differences that affect both threat perceptions and policy preferences (Bullock, Green and Ha, 2010; Imai et al., 2011). Second, our design includes three potential mediators measured simultaneously, raising concerns about causal ordering among mediators and whether mediators are themselves confounded (VanderWeele, 2014). The theoretical pathway in CAM suggests a sequence (information $\rightarrow$ updated beliefs $\rightarrow$ affective response $\rightarrow$ policy demand), but we cannot distinguish whether AutoAnxiety (cognitive) precedes or follows EgoThreat (affective). Future designs should consider temporal separation of mediator measurements or experimental manipulation of the mediators themselves (Pirlott and MacKinnon, 2016; Spencer, Zanna and Fong, 2005).

Despite these limitations, two findings provide suggestive support for the CAM mediational logic. The treatment successfully activates threat-related cognitions and emotions (the "first stages" are significant for 2 of 3 mediators), and the proportion mediated (14.8%) is consistent with partial mediation—implying that most of the treatment effect operates through channels we did not measure, potentially including the moderation pathways documented above or alternative mechanisms such as identity threat or social learning. Full results tables are reported in Appendix D.

6 Discussion

6.1 Summary of Findings

The unconditional ATE of personalized automation risk information on policy demand is positive and robust across estimators ($d \approx 0.23$–0.28), but this aggregate estimate masks the heterogeneity that CAM predicts.

[8] AutoAnxiety (Q94) items: "How much do you agree or disagree with the following statements about your job and automation?" (1) "My job is likely to be automated in the next 5 years"; (2) "My job is likely to be automated in the next 10 years"; (3) "I am concerned about losing my job to automation or AI"; (4) "I feel secure that my occupation is safe from automation" (reverse-coded). All items on 1–7 agreement scale. EgoThreat (Q95, items 1, 3, 5, 7): "To what extent do you believe that advances in AI will have the following impacts on your career and job prospects?" (1) "Reduced job opportunities: AI will lead to fewer available jobs for me, making it harder to find a job"; (3) "Lower salary and compensation: AI will cause a decline in my future wages and benefits in my field"; (5) "Decreased job security: AI will make my employment less stable or increase my risk of being laid off"; (7) "Worsened work conditions: AI will lead to a decline in my pay, benefits, or the quality of my work environment." EgoOpp (Q95, items 2, 4, 6): (2) "New jobs: AI-driven innovation will spur economic growth and create new job opportunities for me"; (4) "Enhanced job satisfaction: AI's automation of routine tasks will free up my time for creative and strategic work"; (6) "Enhanced productivity: AI will provide tools and assistance that allow me to complete my work more efficiently and effectively." All Q95 items on 1–5 scale (Definitely False to Definitely True).

Approximately 62% of respondents are classified as Skeptics by the latent trust mixture, and their treatment effects are statistically indistinguishable from zero across the full range of both the Belief Gap and trait anxiety—confirming *Gate 1* as a strict constraint on information processing. Among Trusters who clear the first gate, treatment effects are not automatic: an activation threshold emerges at approximately the 20th percentile of trait anxiety, below which even trusted, expectation-violating information fails to shift policy preferences. The maximum conditional effect—$d = 0.61$ (95% CI: [0.40, 0.86]) at High Gap $\times$ High Anxiety among Trusters—is more than double the unconditional ATE, providing direct evidence for the two-gate activation logic at the core of CAM.

6.2 Implications for the Automation Literature

The null aggregate effects reported in several recent automation experiments—Zhang (2022), Magistro et al. (2024), Ladreit (2022)—are consistent with the ATE dilution mechanism developed in Section 2. Each study confirmed that respondents received and processed the informational treatment (via manipulation checks or belief updating), yet policy preferences remained unchanged in aggregate. Our results suggest a specific account of this pattern: if these samples contained Skeptic proportions comparable to the 62% we estimate, then genuine persuasion among responsive Trusters would be diluted below conventional significance thresholds when pooled with the null responses of the majority. The implication is that prior null findings may reflect a composition effect—heterogeneous subgroups canceling in aggregation—rather than genuine preference stability in the face of expert information.

Studies that modeled heterogeneity explicitly report patterns consistent with CAM's gating structure. Arntz, Gregory and Zierahn (2024) found opposing treatment effects by prior beliefs—technological optimists reduced policy demand while neutral respondents increased it—producing null effects only in aggregate, precisely the cancellation pattern that motivates our latent class decomposition. Heinrich and Witko (2024) found policy effects exclusively among respondents with high prior technology knowledge, a construct that plausibly correlates with openness to expert information and thus with clearing *Gate 1*. Shen et al. (2025) found that generative-AI information shifted social policy preferences only among prior AI users, again suggesting that baseline familiarity—and the trust in expert forecasts it may engender—conditions responsiveness. Jeffrey (2023) showed that fairness framing mediated treatment effects on redistribution preferences, consistent with *Gate 2*: affective engagement (here, perceived unfairness) was necessary to translate accepted information into policy demand. Across these studies, the common thread is that treatment effects emerge only when conditions analogous to one or both of CAM's gates are satisfied.

CAM predicts that the two gates activate policy *search*, not policy *direction*—and several findings underscore this distinction. Wu (2023) document partisan divergence in preferred policy responses to automation threat, Gallego and Kurer (2022) find that threatened workers favor protectionism over redistribution, and Lee (2024) shows that priming distributive consequences shapes whether respondents support restriction or compensation, with institutional context (UK vs. Denmark) further moderating the direction of demand. Our policy battery, which emphasizes government-led interventions such as retraining, social spending, and regulation, captures demands that align more naturally with left-leaning preferences—an acknowledged scope condition that may lead us to underestimate treatment effects among right-leaning respondents whose policy search channels into market-based or restrictionist solutions. More broadly, the cross-national patterns reported by Lee (2024) and Shen et al. (2025)—where institutional safety nets and sector of employment shape which policies gain support—suggest that CAM's gates operate within an institutional envelope that our U.S.-only design cannot test.

6.3 Theoretical Contributions

The persuasion literature typically treats source credibility as a continuous moderator that adjusts the slope of treatment effects—more trust yields stronger persuasion, less trust yields weaker persuasion (Lupia and McCubbins, 1998; Druckman, 2001). Our latent class results suggest something qualitatively different: Skeptics show zero treatment effects across the full range of both moderators, not merely attenuated effects. This pattern is more consistent with trust operating as a necessary condition—a discrete gate that either permits or blocks informa-

tion processing—than as a magnitude modifier. The distinction connects *Gate 1* to the acceptance filter in Zaller (1992)'s receive-accept-sample model: information that fails the acceptance screen is not merely downweighted but excluded from the considerations that shape preference formation.

A further complication is that the gate itself is not fixed prior to the informational encounter. Our measurement model separates dispositional trust from the epistemic validation effect of the Belief Gap (via Gap and Gap^2 covariates), revealing that large discrepancies between expert and prior assessments erode source credibility upon exposure. An individual who enters the experiment trusting experts may cross below the gate threshold when the message sharply contradicts prior beliefs—the informational content of the vignette can itself induce source rejection. This means that even a pre-treatment measure of expert trust would not fully resolve the identification problem, because the operationally relevant trust state is the one that obtains *after* the dissonant information has been processed. The implication for the broader persuasion literature is that source credibility and message content are not independent inputs to attitude change; rather, the credibility of the source is partly a function of how challenging the message is to the recipient's prior beliefs (Taber and Lodge, 2006). Designs that seek to model trust-moderated effects should therefore account for this endogeneity, whether through the kind of collider-blocking strategy we employ or through experimental manipulation of source credibility orthogonal to message content.

Trait anxiety amplifies treatment effects among Trusters but has no detectable effect among Skeptics—a pattern that discriminates between competing interpretations of anxiety's role in political information processing. If anxiety operated as simple worry, it should increase responsiveness regardless of source acceptance: worried individuals would react more strongly to threatening information whether or not they trusted its source. Instead, anxiety functions as AIT predicts (Marcus, Neuman and MacKuen, 2000; Valentino et al., 2008): it sustains accuracy-motivated processing of information that has already cleared the acceptance filter, shifting cognition from habitual reliance on standing dispositions toward deliberative evaluation. The activation threshold at approximately the 20th percentile of trait anxiety further sharpens this interpretation. Below this threshold, even Trusters who receive large, expectation-violating Belief Gaps show null treatment effects—a "cool cognition" pattern in which belief updating may occur without generating the urgency needed to revise policy preferences. The threshold suggests that a minimum level of dispositional vigilance is necessary for the surveillance system to sustain elaboration long enough to reach downstream attitudes, consistent with Albertson and Gadarian (2015)'s finding that anxious citizens increase reliance on trusted experts rather than abandoning all heuristics.

The observed heterogeneity maps onto a three-pathway taxonomy that distinguishes the cognitive depth of treatment responses. Skeptics exhibit a modest baseline positive effect at the intercept ($d = 0.23$) that does not scale with either the Belief Gap or trait anxiety—a pattern consistent with shallow priming, in which the treatment temporarily raises the salience of automation as a policy-relevant issue without engaging deeper processing (Druckman et al., 2022). Among low-anxiety Trusters, a different mechanism produces a similar floor: despite accepting the source and receiving large Belief Gaps, these respondents show treatment effects that converge at the baseline rather than scaling with the informational shock. This "cool cognition" pattern suggests belief updating without urgency—the information is registered but does not activate the policy search that CAM requires at *Gate 2*. Only among high-anxiety Trusters with positive Gaps do treatment effects scale with both moderators, consistent with deep persuasion in which accuracy-motivated processing translates accepted, threatening information into revised policy preferences. These three pathways—source rejection, shallow priming, and cool cognition without activation—offer a more differentiated account of null aggregate effects than the binary distinction between "information works" and "preferences are stable."

6.4 Methodological Implications

A central identification challenge in our design—and in any study where source credibility conditions treatment effects—is that trust can only be observed for treated units. Asking control respondents about the expert source's credibility would reveal the treatment and contaminate the control condition. Our Bayesian finite mixture model addresses this through principal stratification logic (Frangakis and Rubin, 2002): because treatment is randomized, the prevalence of latent trust types is identical across arms, and the estimated mixing proportion π can be

applied to classify control units at the population base rate. A further complication is that the observed trust indicator Q_{obs} is a collider: both the Belief Gap and dispositional trust causally influence trust responses (Gap $\rightarrow Q_{\text{obs}} \leftarrow S$), so naive conditioning on observed trust conflates dispositional acceptance with epistemic validation. Our measurement model blocks this backdoor path by including Gap and Gap^2 as covariates, isolating the component of trust that is orthogonal to informational content. The technical details of this identification strategy are developed in the model specification (Section 3); the implication for the broader literature is that post-treatment moderators require structural modeling rather than sample splitting or regression interaction, both of which inherit the collider bias.

To illustrate the practical consequences of ignoring the collider structure, consider a naive analysis that splits the sample by observed trust (e.g., respondents reporting $Q_{\text{obs}} \geq 4$ classified as "high trust"). Because trust responses are partly driven by the Belief Gap—respondents whose priors closely match the expert estimate report higher trust—selecting on high observed trust systematically oversamples individuals with small Gaps. These are precisely the respondents for whom the treatment conveys little new information, yielding small treatment effects by construction. The resulting estimate of "trust-moderated persuasion" would be attenuated not because trust fails to amplify effects, but because the selection criterion screens out the high-Gap individuals who would show the strongest responses. Conversely, the "low trust" subsample would overrepresent respondents with large Gaps—individuals who may have entered the experiment trusting experts but were reclassified by epistemic dissonance. The analyst would conclude, incorrectly, that trust moderation is weak, when in fact the conditioning strategy has mixed dispositional Skeptics with dissonance-driven rejectors and excluded the most informative Trusters. Our mixture model avoids this trap by estimating dispositional trust net of the Gap, recovering the theoretically relevant moderator rather than its confounded observable proxy.

6.5 Limitations

Our sample consists of $N = 910$ U.S. workers recruited via CloudResearch Connect. The sample skews liberal (approximately 62% Democrat), and while it includes meaningful conservative representation, generalizing to the broader U.S. workforce or to other national contexts requires caution. Baseline trust in experts varies substantially across countries (Gauchat, 2012), and the prevalence of Trusters and Skeptics in our latent class model may not be representative beyond U.S. online workers. Observed trust in our expert source is correlated with partisanship and ideology: among treated respondents, high-trust individuals ($Q_{\text{obs}} \geq 4$) are disproportionately Democratic and liberal relative to low-trust respondents. Because left-leaning respondents are also more predisposed to support government-led policy interventions, trust-moderated treatment effects could partly reflect partisan sorting rather than dispositional trust per se. Our structural model includes ideology and partisanship as controls in the policy demand equation, absorbing baseline partisan differences in the outcome and ensuring that treatment effect estimates are not confounded by ideological predisposition.[9]

Trust is measured with a single 5-point ordinal item assessing confidence in the Oxford researchers' competency to evaluate automation risk for the respondent's specific occupation. We cannot determine whether this item captures a stable dispositional tendency to trust expert authority—as CAM theorizes—or a situation-specific credibility judgment about these particular researchers in this domain. Our mixture model mitigates this concern in part by controlling for prior exposure to automation risk information (in both linear and quadratic form), separating dispositional trust from familiarity-driven credibility; nonetheless, a multi-item battery spanning source-specific and generalized expert trust would clarify whether *Gate 1* reflects a stable disposition or a context-dependent evaluation, and experimental manipulation of source credibility would provide exogenous variation in trust.

The mixture model introduces classification uncertainty that propagates into treatment effect estimates. The separation parameter $\delta = 2.09$ has wide credible intervals (95% CI: [0.30, 3.95]), and the Truster prevalence

[9] A more targeted approach—including ideology and partisanship as covariates in the trust measurement model itself—would estimate the trust class separation δ conditional on partisan identity, directly severing the sorting channel. The liberal skew of our sample limits the statistical power for this specification; a more ideologically balanced sample would provide the variation needed to disentangle dispositional trust from partisan sorting directly.

$\pi = 0.381$ spans [0.12, 0.75]. The qualitative pattern—larger effects among Trusters, amplification at High Gap $\times$ High Anxiety—is robust across the posterior range, but the precise magnitude of class-specific effects shifts meaningfully with different π values.

The expert risk estimates derive from Frey and Osborne's (2017) automation probabilities, computed based on technological capabilities circa 2013–2017. Rapid advances in generative AI may have changed the risk landscape, and these estimates operationalize a specific conception of automation risk—task-level susceptibility to computerization—that may not capture augmentation, reorganization, or other dimensions of technological displacement.

The policy battery emphasizes government-led interventions (retraining, social spending, regulation) that align with progressive policy preferences. This may underestimate treatment effects among right-leaning respondents whose activated policy search channels into market-based or restrictionist solutions. Ideologically balanced batteries capturing both the magnitude and direction of policy response are needed.

We measure policy preferences immediately post-treatment and cannot assess whether effects persist, decay, or strengthen over time. The distinction between transient accessibility effects and durable attitude change is critical for evaluating CAM's practical implications for risk communication.

Finally, our three-step belief correction protocol—eliciting prior beliefs, presenting personalized expert feedback, and asking respondents to evaluate the discrepancy—is more intensive than naturalistic information encounters. While this design choice was motivated by the need to ensure sufficient engagement depth at *Gate 2*, it limits ecological validity: effects may not transfer to settings where citizens encounter automation risk information passively or in less structured formats.

6.6 Future Directions

Several extensions would strengthen and generalize the CAM framework. First, experimentally manipulating source credibility—varying whether automation risk information is attributed to academic experts, government agencies, or industry sources—would provide exogenous variation in *Gate 1* and allow cleaner causal identification of trust-moderated effects than our observational latent class approach. Second, panel designs with pre-treatment, immediate post-treatment, and delayed waves would clarify whether the conditional effects we document reflect durable attitude formation or transient accessibility, a distinction our single-wave design cannot adjudicate. Third, replicating CAM in national contexts with higher or lower baseline expert trust would test whether the two-gate structure is generalizable or culturally contingent; the prevalence of Trusters and Skeptics may vary even if the conditional logic remains constant. Fourth, ideologically balanced policy batteries—spanning government intervention, market-based solutions, and restrictionist measures—would capture the full heterogeneity of policy response direction that our left-leaning battery likely understates. Finally, the theoretical logic of CAM—that information effects are conditional on source acceptance and affective activation—is not specific to automation risk. Climate change, vaccine efficacy, and trade policy present analogous structures of contested expertise and identity relevance, and testing whether the two-gate architecture applies in these domains would establish whether CAM identifies a general mechanism of conditional persuasion or a domain-specific pattern.

7 Conclusion

When does expert information about AI-driven automation risk translate into public demand for protective labor policy? Experimental studies that expose respondents to information about how automation and artificial intelligence may displace jobs have produced a mixed pattern: several report null effects on policy preferences despite successful belief updating, while others find conditional effects that emerge only among specific subgroups. This paper proposes the Conditional Activation Model to organize these findings and tests it in a pre-registered survey experiment with $N = 910$ U.S. workers. CAM formalizes two sequential gates through which expert information must pass before it generates policy demand: source trust (*Gate 1*) and the joint activation of a

positive Belief Gap and trait anxiety (*Gate 2*).

The core empirical results are robust across estimation strategies. The unconditional ATE ($d \approx 0.23$–0.28) is positive and consistent across OLS, SEM, and the Bayesian finite mixture model, confirming that personalized automation risk information shifts policy preferences on average. The qualitative heterogeneity pattern—zero effects among Skeptics, an activation threshold at the 20th percentile of trait anxiety among Trusters, and maximum effects of $d = 0.61$ at High Gap $\times$ High Anxiety—holds across the posterior distribution despite substantial classification uncertainty ($\pi \in [0.12, 0.75]$). These results are not, however, without fragility: they rest on a single experiment with a liberally skewed sample, a one-item trust measure whose relationship to stable dispositional trust remains uncertain, and a mixture model whose identifying assumptions—particularly the independence of dispositional trust from prior beliefs—are untestable.

Two contributions merit emphasis. First, the finding that trust operates as a discrete gate rather than a continuous moderator—Skeptics show zero effects, not merely attenuated ones—has implications beyond the automation domain. Any persuasion study in which source acceptance varies across respondents risks ATE dilution if heterogeneity is unmodeled, and the collider structure that arises when trust is post-treatment (Gap $\rightarrow Q_{\text{obs}} \leftarrow S$) requires structural approaches rather than naive conditioning. Second, the three-pathway decomposition—source rejection, shallow priming, and deep persuasion—provides a more differentiated account of null aggregate effects than the binary framing of "information works" versus "preferences are stable" that has characterized much of this literature.

The translation of expert information into public policy demand is neither automatic nor impossible—it is conditional on psychological gates that most experimental designs have not been equipped to detect. Whether CAM's two-gate architecture generalizes beyond automation risk to other domains of contested expertise—climate, public health, trade—remains an open empirical question that future work should address with more balanced samples, exogenous trust manipulations, and panel designs capable of distinguishing transient accessibility from durable attitude change.

A Bayesian Finite Mixture Model: Full Specification

This appendix provides the complete specification of our Bayesian Finite Mixture Model (BFMM), including all measurement models, the structural model, prior distributions, and identification restrictions. The model is estimated using Stan via Hamiltonian Monte Carlo (HMC) with the No-U-Turn Sampler (NUTS).

A.1 Model Overview

The BFMM integrates three components:

1. **Trust Mixture Model (Gate 1)**: A two-component mixture classifying respondents as Trusters ($S_i = 1$) or Skeptics ($S_i = 0$) based on their ordinal trust response.
2. **Measurement Models**: Confirmatory factor analysis (CFA) models for trait anxiety (predictor) and policy demand (outcome).
3. **Structural Model (Gate 2)**: A regression of latent policy demand on treatment, the Belief Gap, latent trait anxiety, and their interactions—with class-specific effects.

A.2 Trust Mixture Model (Gate 1)

The trust measurement model classifies treated respondents into latent classes based on their observed trust response $Q_i \in \{1, 2, 3, 4, 5\}$.

A.2.1 Latent Class Assignment

Each treated individual belongs to a latent trust class S_i:

$$S_i \sim \text{Bernoulli}(\pi)$$

where π is the population prevalence of Trusters. We estimate π as a single scalar parameter (constant across individuals).

A.2.2 Trust Measurement Model

The observed ordinal trust response follows an ordered logistic model with class-specific intercepts:

$$Q_i \mid S_i, \boldsymbol{X}_{\text{trust},i} \sim \text{OrderedLogit}\left(\delta \cdot S_i + \boldsymbol{X}'_{\text{trust},i}\boldsymbol{\gamma}_{\text{trust}}, \boldsymbol{\kappa}\right)$$

where:

- $\delta > 0$ is the *separation parameter*—the log-odds shift in trust for Trusters relative to Skeptics
- $\boldsymbol{X}_{\text{trust},i}$ is a vector of trust covariates (Gap, Gap^2, Prior Exposure, Prior Exposure^2)
- $\boldsymbol{\gamma}_{\text{trust}}$ are the covariate effects on the trust response
- $\boldsymbol{\kappa} = (\kappa_1, \kappa_2, \kappa_3, \kappa_4)'$ are ordered cutpoints ($\kappa_1 < \kappa_2 < \kappa_3 < \kappa_4$)

A.2.3 Mixture Likelihood

Since the latent class S_i is unobserved, we marginalize over both classes:

$$P(Q_i = k \mid \boldsymbol{X}_{\text{trust},i}) = \pi \cdot P(Q_i = k \mid S_i = 1) + (1 - \pi) \cdot P(Q_i = k \mid S_i = 0)$$

In log-probability form (for numerical stability):

$$\log P(Q_i) = \text{log_sum_exp}\left[\log \pi + \log P(Q_i \mid S_i = 1),\ \log(1 - \pi) + \log P(Q_i \mid S_i = 0)\right]$$

A.2.4 Trust Covariates: Epistemic Validation

The trust covariates capture systematic variation in trust response beyond latent class membership. Based on preliminary analysis, we include:

- **Gap** and **Gap**^{2}: Captures an inverted-U relationship where trust peaks when expert estimates confirm prior beliefs (Gap ≈ 0) and declines as the discrepancy increases in either direction (*epistemic validation*).
- **Prior Exposure** and **Prior Exposure**^{2}: Captures a U-shaped relationship where both naive (low exposure) and sophisticated (high exposure) individuals report higher trust, while those with moderate exposure are most skeptical.

A.3 Trait Anxiety Measurement Model (Predictor)

Latent trait anxiety $\eta_{\text{trait},i}$ is measured via a single-factor CFA using four Big Five-style items (Q23_1 through Q23_4). Items enter the model on the **raw 1–7 Likert scale** (not pre-standardized), which requires estimating item intercepts:

$$Y_{\text{trait},ij} = \nu_{\text{trait},j} + \lambda_{\text{trait},j} \cdot \eta_{\text{trait},i} + \varepsilon_{\text{trait},ij}$$

where:

- $Y_{\text{trait},ij}$ is respondent i's raw response (1–7) to item j
- $\nu_{\text{trait},j}$ is the item intercept (prior: $\mathcal{N}(4, 2)$, centered on scale midpoint)
- $\lambda_{\text{trait},j} > 0$ is the factor loading (positive constrained)
- $\varepsilon_{\text{trait},ij} \sim \mathcal{N}(0, 1)$ is the residual (**fixed** for identification)
- $\eta_{\text{trait},i}$ is the latent factor with freely estimated variance

Identification (Reference Indicator Approach): We fix the first factor loading $\lambda_{\text{trait},1} = 1$ and the residual variance $\sigma^2_{\text{trait}} = 1$. This *reference indicator* approach sets the latent factor scale to match the metric of the first item (Q23_1). The remaining loadings $\lambda_{\text{trait},j}$ for $j = 2, \ldots, 4$ are freely estimated with prior $\mathcal{N}^+(0.8, 0.3)$. Item intercepts $\nu_{\text{trait},j}$ are freely estimated with prior $\mathcal{N}(4, 2)$.

Why Raw Scale: Using raw 1–7 items avoids an identification problem. With pre-standardized items ($\text{Var}(Y) = 1$), fixing both $\lambda_1 = 1$ and $\sigma^2 = 1$ forces $\text{Var}(\eta) = 0$ via the variance decomposition $\text{Var}(Y) = \lambda^2 \text{Var}(\eta) + \sigma^2$. With raw items ($\text{Var}(Y) \approx 2^2$), realistic latent variance is estimable.

Item Reliability: Under this specification, the reliability of item j is:

$$\rho_j = \frac{\lambda_j^2 \cdot \text{Var}(\eta_{\text{trait}})}{\lambda_j^2 \cdot \text{Var}(\eta_{\text{trait}}) + 1}$$

A.4 Policy Demand Measurement Model (Outcome)

Latent policy demand $\eta_{\text{policy},i}$ is measured via a single-factor CFA using five policy support items (Q96_1 through Q96_5). Items are on the **raw 1–7 Likert scale** (not pre-standardized), which requires estimating item intercepts:

$$Y_{\text{policy},ij} = \alpha_{\text{policy},j} + \lambda_{\text{policy},j} \cdot \eta_{\text{policy},i} + \varepsilon_{\text{policy},ij}$$

where:

- $Y_{\text{policy},ij}$ is respondent i's raw response (1–7) to item j
- $\alpha_{\text{policy},j}$ is the item intercept (prior: $\mathcal{N}(4, 2)$, centered on scale midpoint)

- $\lambda_{\text{policy},j} > 0$ is the factor loading (positive constrained)
- $\varepsilon_{\text{policy},ij} \sim \mathcal{N}(0,1)$ is the residual (**fixed** for identification)

Why Raw Scale: Using raw 1–7 items avoids an identification problem. With pre-standardized items ($\text{Var}(Y) = 1$), fixing both $\lambda_1 = 1$ and $\sigma^2 = 1$ forces $\text{Var}(\eta) = 0$ via the variance decomposition $\text{Var}(Y) = \lambda^2\text{Var}(\eta) + \sigma^2$. With raw items ($\text{Var}(Y) \approx 2.5^2$), realistic latent variance is possible.

Identification (Reference Indicator Approach): We fix the first factor loading $\lambda_{\text{policy},1} = 1$ and the residual variance $\sigma^2_{\text{policy}} = 1$. This *reference indicator* approach sets the latent factor scale to match the metric of the first item (Q96_1). The remaining loadings $\lambda_{\text{policy},j}$ for $j = 2, \ldots, 5$ are freely estimated with prior $\mathcal{N}^+(0.8, 0.3)$. Item intercepts $\alpha_{\text{policy},j}$ are freely estimated with prior $\mathcal{N}(4, 2)$.

Key Insight: By fixing $\lambda_1 = 1$, treatment effects ($\boldsymbol{\beta}$) are expressed on the scale of the first policy item. To report effects in SD units of the latent outcome, we standardize by $\text{SD}(\eta_{\text{policy}})$ in post-processing.

Converting to Observed Scale: Treatment effects on latent policy demand can be converted to the observed item scale by multiplying by the average factor loading:

$$\text{CATE}_{\text{observed}} = \text{CATE}_{\text{latent}} \times \bar{\lambda}_{\text{policy}}$$

Item Reliability: Under this specification:

$$\rho_j = \frac{\lambda_j^2 \cdot \text{Var}(\eta_{\text{policy}})}{\lambda_j^2 \cdot \text{Var}(\eta_{\text{policy}}) + 1}$$

A.5 Structural Model (Gate 2)

The structural model specifies how latent policy demand depends on treatment assignment, the Belief Gap, latent trait anxiety, and control variables—with *class-specific* treatment effects.

A.5.1 Canonical SEM Specification

For each individual i, the latent policy demand factor follows:

$$\eta_{\text{policy},i} = \mu_{\text{policy},i} + \sigma_{\text{struct}} \cdot \zeta_i, \quad \zeta_i \sim \mathcal{N}(0,1)$$

where:

- $\mu_{\text{policy},i}$ is the *structural mean* (explained by treatment, covariates, and class membership)
- ζ_i is the standardized structural residual
- $\sigma_{\text{struct}} > 0$ is the structural residual SD (**estimated**, prior: $\text{Exp}(0.5)$, $E[\sigma_{\text{struct}}] = 2$)

This *canonical SEM* parameterization separates:

- **Structural variance**: $\text{Var}(\mu_{\text{policy}})$ = variance explained by treatment + covariates
- **Structural residual variance**: σ^2_{struct} = unexplained latent variance
- **Measurement error variance**: $\sigma^2_{\text{policy}} = 1$ (fixed)

The structural R^2 is:

$$R^2_{\text{structural}} = \frac{\text{Var}(\mu_{\text{policy}})}{\text{Var}(\mu_{\text{policy}}) + \sigma^2_{\text{struct}}}$$

A.5.2 Mixture Structure

The structural mean is a mixture over latent trust classes:

$$\mu_{\text{policy},i} = \pi \cdot \mu_{\text{truster},i} + (1 - \pi) \cdot \mu_{\text{skeptic},i}$$

A.5.3 Class-Specific Structural Means

For Skeptics ($S_i = 0$):

$$\mu_{\text{skeptic},i} = \alpha_{\text{skeptic}} + T_i \cdot \tau_{\text{skeptic},i} + \boldsymbol{X}'_{\text{ctrl},i}\boldsymbol{\beta}_{\text{ctrl}}$$

For Trusters ($S_i = 1$):

$$\mu_{\text{truster},i} = (\alpha_{\text{skeptic}} + \alpha_{\Delta}) + T_i \cdot \tau_{\text{truster},i} + \boldsymbol{X}'_{\text{ctrl},i}\boldsymbol{\beta}_{\text{ctrl}}$$

where:

- $T_i \in \{0, 1\}$ is the treatment indicator
- α_{skeptic} is the baseline policy demand for Skeptics (control group mean)
- $\alpha_{\Delta} \geq -0.1$ is the baseline difference for Trusters (lower-bounded to allow near-zero values)
- $\boldsymbol{X}_{\text{ctrl},i}$ is a vector of pre-treatment control variables

A.5.4 Conditional Average Treatment Effects (CATE)

The treatment effect functions τ_s are specified as:

Skeptics:

$$\tau_{\text{skeptic},i} = \beta_0^{(S)} + \beta_1^{(S)} \cdot \text{Gap}_i + \beta_2^{(S)} \cdot \eta_{\text{trait},i} + \beta_3^{(S)} \cdot (\text{Gap}_i \times \eta_{\text{trait},i})$$

Trusters:

$$\tau_{\text{truster},i} = \beta_0^{(T)} + \beta_1^{(T)} \cdot \text{Gap}_i + \beta_2^{(T)} \cdot \eta_{\text{trait},i} + \beta_3^{(T)} \cdot (\text{Gap}_i \times \eta_{\text{trait},i})$$

We parameterize the Truster coefficients as additive shifts from the Skeptic coefficients:

$$\boldsymbol{\beta}^{(T)} = \boldsymbol{\beta}^{(S)} + \boldsymbol{\beta}_{\Delta}, \quad \text{where } \boldsymbol{\beta}_{\Delta} \geq 0$$

The constraint $\boldsymbol{\beta}_{\Delta} \geq 0$ encodes the theoretical expectation that Trusters exhibit *weakly larger* treatment effects (they do not discount the expert information).

A.5.5 Key Estimand: The CATE Surface

The core theoretical quantity is the full CATE surface $\tau_{\text{truster}}(\text{Gap}, \eta_{\text{trait}})$, evaluated across a grid of Belief Gap and trait anxiety values. The Conditional Activation Model predicts that treatment effects among Trusters should be (1) positive in expectation, (2) increasing in Gap (larger informational shocks produce stronger responses), and (3) increasing in trait anxiety (dispositional vigilance amplifies the response). We assess these predictions by computing posterior CATE estimates at Low/Mean/High (± 1 SD) combinations of both moderators and examining whether the surface exhibits the predicted gradient.

A.6 Prior Distributions

All priors are weakly informative, centered on theoretically plausible values while allowing the data to dominate inference.

Component	Parameter	Prior	Rationale
Trust Model	π	$\text{Beta}(2, 2)$	Weakly informative, centered at 0.5
	δ	$\mathcal{N}^+(3, 3)$	Positive separation; expects moderate effect
	$\boldsymbol{\kappa}$	$\mathcal{N}(0, 1.5)$	Regularized cutpoints
	$\boldsymbol{\gamma}_{\text{trust}}$	$\mathcal{N}(0, 1)$	Regularized covariate effects
Trait CFA	$\boldsymbol{\nu}_{\text{trait}}$	$\mathcal{N}(3, 1.5)$	Item intercepts (raw 1–7 scale)
	$\lambda_{\text{trait},1}$	$= 1$ (fixed)	Reference indicator
	$\boldsymbol{\lambda}_{\text{trait},2:J}$	$\mathcal{N}^+(0.9, 0.3)$	Positive loadings near unity
	$\eta_{\text{trait},i}$	$\mathcal{N}(0, 1)$	Soft prior (scale stabilization)
Policy CFA	$\boldsymbol{\alpha}_{\text{policy}}$	$\mathcal{N}(4, 2)$	Item intercepts (raw 1–7 scale)
	$\lambda_{\text{policy},1}$	$= 1$ (fixed)	Reference indicator
	$\boldsymbol{\lambda}_{\text{policy},2:J}$	$\mathcal{N}^+(0.8, 0.3)$	Positive loadings, moderate values
Structural	α_{skeptic}	$\mathcal{N}(0, 1)$	Regularized baseline
	α_Δ	$\mathcal{N}(0, 0.5)$, $\alpha_\Delta \geq -0.1$	Tight prior; allows near-zero separation
	$\boldsymbol{\beta}^{(S)}$	$\mathcal{N}(0, 0.5)$	Regularized for realistic effect sizes
	$\boldsymbol{\beta}_\Delta$	$\mathcal{N}^+(0, 0.25)$	Non-negative, tight: 95% < 0.5
	$\boldsymbol{\beta}_{\text{ctrl}}$	$\mathcal{N}(0, 0.5)$	Regularized control effects
	σ_{struct}	$\text{LogNormal}(0.4, 0.5)$	Median ≈ 1.5; 95% CI $\approx (0.6, 3.7)$
	ζ_i	$\mathcal{N}(0, 1)$	Standardized structural residual

Table 4: Prior distributions for all model parameters. $\mathcal{N}^+$ denotes a half-normal (truncated at 0) or lower-bounded normal distribution. Both trait anxiety items (raw 1–7 scale) and policy items (raw 1–7 scale) enter the model unstandardized, requiring intercept estimation. The first factor loading ($\lambda_1 = 1$) is fixed for both measurement models to set the latent scale.

A.7 Identification Restrictions

The model is identified through the following restrictions:

1. **Trait Anxiety Factor Scale (Reference Indicator)**: $\lambda_{\text{trait},1} = 1$ (fixed). This sets the latent factor scale to match the metric of the first trait anxiety item ("I see myself as anxious, easily upset").
2. **Trait Measurement Error Variance**: $\sigma^2_{\varepsilon,\text{trait}} = 1$ (fixed). This fixes the measurement error variance (not the latent factor variance), providing a second identification constraint for the trait CFA.
3. **Policy Factor Scale (Reference Indicator)**: $\lambda_{\text{policy},1} = 1$ (fixed). This sets the latent factor scale to match the metric of the first policy item.
4. **Policy Measurement Error Variance**: $\sigma^2_{\varepsilon,\text{policy}} = 1$ (fixed). This fixes the measurement error variance (not the latent factor variance), providing a second identification constraint for the policy CFA.
5. **Trust Class Separation**: $\delta > 0$ ensures Trusters report systematically higher trust than Skeptics, breaking label-switching symmetry.
6. **Treatment Effect Direction**: $\boldsymbol{\beta}_\Delta \geq 0$ encodes the theoretical constraint that Trusters exhibit weakly larger treatment effects.
7. **Mixture Prevalence**: $\pi \in (0, 1)$ estimated as a single scalar (homogeneous across individuals).
8. **Raw Scale Data**: Both trait anxiety items (Y_{trait}) and policy items (Y_{policy}) enter the model on their raw 1–7 Likert scales (not pre-standardized). Item intercepts ν_j are estimated for both measurement models. This ensures proper identification under the reference indicator approach ($\lambda_1 = 1$, $\sigma^2_\varepsilon = 1$).

A.8 Derived Quantities

A.8.1 Implied Truster Coefficients

$$\boldsymbol{\beta}^{(T)} = \boldsymbol{\beta}^{(S)} + \boldsymbol{\beta}_{\Delta}$$

A.8.2 Standardized Treatment Effect Coefficients

Since η_{trait} may have variance $\neq 1$ (soft prior), we re-express the CATE in terms of standardized predictors. Let $\bar{\eta}_{\text{trait}}$ and $s_{\eta_{\text{trait}}}$ denote the posterior mean and SD of the latent trait factor. Then:

$$\boldsymbol{\beta}^{(T)}_{\text{adj}} = \begin{pmatrix} \beta_0^{(T)} + \beta_2^{(T)} \bar{\eta}_{\text{trait}} \\ \beta_1^{(T)} + \beta_3^{(T)} \bar{\eta}_{\text{trait}} \\ \beta_2^{(T)} \cdot s_{\eta_{\text{trait}}} \\ \beta_3^{(T)} \cdot s_{\eta_{\text{trait}}} \end{pmatrix}$$

These adjusted coefficients give effects per 1 SD change in both Gap (already standardized in input) and trait anxiety.

A.8.3 CATE Grid (Latent Scale)

We compute treatment effects at a 3×3 grid of Gap × Anxiety values (Low/Mean/High = $-1/0/+1$ SD):

$$\text{CATE}(g, a) = \beta^{(T)}_{0,\text{adj}} + \beta^{(T)}_{1,\text{adj}} \cdot g + \beta^{(T)}_{2,\text{adj}} \cdot a + \beta^{(T)}_{3,\text{adj}} \cdot (g \times a)$$

where $g, a \in \{-1, 0, 1\}$ represent standardized Gap and Anxiety values.

A.8.4 CATE Grid (SD Units of Outcome)

To express effects in SD units of the latent outcome:

$$\text{CATE}_{\text{SD}}(g, a) = \frac{\text{CATE}(g, a)}{\text{SD}(\eta_{\text{policy}})}$$

where $\text{SD}(\eta_{\text{policy}}) = \sqrt{\text{Var}(\mu_{\text{policy}}) + \sigma^2_{\text{struct}}}$.

A.8.5 CATE Grid (Observed Scale)

Treatment effects converted to the original item scale (before standardization):

$$\text{CATE}_{\text{obs}}(g, a) = \text{CATE}(g, a) \times \bar{\lambda}_{\text{policy}}$$

where $\bar{\lambda}_{\text{policy}} = \frac{1}{J_{\text{policy}}} \sum_{j=1}^{J_{\text{policy}}} \lambda_{\text{policy},j}$.

A.8.6 Item Reliability

$$\rho_j = \frac{\lambda_j^2 \cdot \text{Var}(\eta)}{\lambda_j^2 \cdot \text{Var}(\eta) + 1}$$

For trait items with $\text{Var}(\eta_{\text{trait}})$ regularized toward 1, this is approximately $\rho_j \approx \lambda_j^2/(\lambda_j^2 + 1)$.

A.8.7 Scale Reliability (Omega)

$$\omega = \frac{\left(\sum_{j=1}^{J} \lambda_j\right)^2 \cdot \text{Var}(\eta)}{\left(\sum_{j=1}^{J} \lambda_j\right)^2 \cdot \text{Var}(\eta) + J}$$

A.8.8 Structural R^2

$$R^2_{\text{structural}} = \frac{\text{Var}(\mu_{\text{policy}})}{\text{Var}(\mu_{\text{policy}}) + \sigma^2_{\text{struct}}}$$

This quantifies the proportion of latent policy demand variance explained by treatment assignment, moderators, and control variables.

A.9 Computation of Key Estimands

This section details the exact formulas used to compute the key quantities of interest from posterior draws. All quantities are computed in the `generated quantities` block of the Stan program, ensuring full propagation of posterior uncertainty.

A.9.1 Class-Specific Treatment Effects at Reference Point

The treatment effect for each trust class, evaluated at the reference point (Gap $= 0$, $\eta_{\text{trait}} = \bar{\eta}_{\text{trait}}$):

Trusters:

$$\tau^{\text{ref}}_{\text{truster}} = \beta_0^{(T)} + \beta_2^{(T)} \cdot \bar{\eta}_{\text{trait}}$$

Skeptics:

$$\tau^{\text{ref}}_{\text{skeptic}} = \beta_0^{(S)} + \beta_2^{(S)} \cdot \bar{\eta}_{\text{trait}}$$

where $\bar{\eta}_{\text{trait}} = \frac{1}{N}\sum_{i=1}^{N} \eta_{\text{trait},i}$ is the posterior mean of the latent trait factor across individuals.

A.9.2 Mixture-Weighted Average Treatment Effect

The unconditional ATE marginalizes over the latent trust class:

At reference point:

$$\text{ATE}^{\text{ref}} = \pi \cdot \tau^{\text{ref}}_{\text{truster}} + (1-\pi) \cdot \tau^{\text{ref}}_{\text{skeptic}}$$

Marginal ATE (averaged over the empirical distribution of Gap and η_{trait}):

$$\text{ATE}^{\text{marg}} = \frac{1}{N}\sum_{i=1}^{N} \left[\pi \cdot \tau_{\text{truster},i} + (1-\pi) \cdot \tau_{\text{skeptic},i}\right]$$

where the individual-level treatment effects are:

$$\tau_{\text{truster},i} = \beta_0^{(T)} + \beta_1^{(T)} \cdot \text{Gap}_i + \beta_2^{(T)} \cdot \eta_{\text{trait},i} + \beta_3^{(T)} \cdot (\text{Gap}_i \times \eta_{\text{trait},i})$$
$$\tau_{\text{skeptic},i} = \beta_0^{(S)} + \beta_1^{(S)} \cdot \text{Gap}_i + \beta_2^{(S)} \cdot \eta_{\text{trait},i} + \beta_3^{(S)} \cdot (\text{Gap}_i \times \eta_{\text{trait},i})$$

A.9.3 Standardization to Cohen's d

All treatment effects are standardized by the SD of the latent outcome to yield effect sizes in Cohen's d units:

$$d = \frac{\tau}{\text{SD}(\eta_{\text{policy}})}, \quad \text{where SD}(\eta_{\text{policy}}) = \sqrt{\text{Var}(\mu_{\text{policy}}) + \sigma^2_{\text{struct}}}$$

A.9.4 CATE Grid Computation

The 3×3 CATE grid evaluates treatment effects for Trusters at all combinations of Low/Mean/High Gap and Anxiety (coded as -1, 0, $+1$ SD):

$$\text{CATE}(g, a) = \beta_{0,\text{std}}^{(T)} + \beta_{1,\text{std}}^{(T)} \cdot g + \beta_{2,\text{std}}^{(T)} \cdot a + \beta_{3,\text{std}}^{(T)} \cdot (g \times a)$$

where the standardized coefficients $\boldsymbol{\beta}_{\text{std}}^{(T)}$ are computed as:

$$\begin{aligned}
\beta_{0,\text{std}}^{(T)} &= \frac{\beta_0^{(T)} + \beta_2^{(T)} \cdot \bar{\eta}_{\text{trait}}}{\text{SD}(\eta_{\text{policy}})} \\
\beta_{1,\text{std}}^{(T)} &= \frac{\beta_1^{(T)} + \beta_3^{(T)} \cdot \bar{\eta}_{\text{trait}}}{\text{SD}(\eta_{\text{policy}})} \\
\beta_{2,\text{std}}^{(T)} &= \frac{\beta_2^{(T)} \cdot s_{\eta_{\text{trait}}}}{\text{SD}(\eta_{\text{policy}})} \\
\beta_{3,\text{std}}^{(T)} &= \frac{\beta_3^{(T)} \cdot s_{\eta_{\text{trait}}}}{\text{SD}(\eta_{\text{policy}})}
\end{aligned}$$

where $s_{\eta_{\text{trait}}} = \sqrt{\text{Var}(\eta_{\text{trait}})}$ is the posterior SD of the latent trait factor.

A.9.5 CATE Contrasts

To test whether the maximum activation condition (High Gap $\times$ High Anxiety) produces larger effects than other conditions, we compute pairwise contrasts:

$$\Delta_{(g,a)} = \text{CATE}(+1, +1) - \text{CATE}(g, a), \quad \text{for all } (g, a) \in \{-1, 0, +1\}^2$$

Posterior probabilities $P(\Delta_{(g,a)} > 0)$ provide evidence for the Conditional Activation hypothesis.

A.10 Estimation

The model is estimated using Stan's No-U-Turn Sampler (NUTS), a variant of Hamiltonian Monte Carlo. We run 8 parallel chains with:

- 2,000 warmup iterations (adaptation)
- 2,000 sampling iterations per chain
- Target acceptance rate: $\delta = 0.95$
- Maximum tree depth: 12

Convergence is assessed via:

- $\hat{R} < 1.01$ for all parameters
- Effective sample size (ESS) > 400 for key parameters
- No divergent transitions post-warmup
- Visual inspection of trace plots and posterior distributions

B Sample Characteristics and Psychometric Validation

This appendix presents detailed sample characteristics, descriptive statistics, and psychometric validation of the multi-item scales used in the analysis.

B.1 Sample Demographics

Table 5: Sample Demographics ($N = 910$)

Variable	Category	n	%
Gender			
	Male	425	46.7
	Female	469	51.5
	Non-binary / Other	11	1.2
	Prefer not to say	5	0.5
Race / Ethnicity			
	White	629	69.1
	Black or African American	83	9.1
	Hispanic or Latino	1	0.1
	Asian	72	7.9
	Native Hawaiian / Pacific Islander	4	0.4
	Other	3	0.3
	Prefer not to say	51	5.6
	Multiracial	67	7.4
Education			
	Less than high school	4	0.4
	High school diploma / GED	99	10.9
	Some college	242	26.6
	Bachelor's degree	400	44.0
	Graduate degree	165	18.1
Party Identification (7-point)			
	Strong Democrat	292	32.1
	Weak Democrat	175	19.2
	Lean Democrat	97	10.7
	Independent	98	10.8
	Lean Republican	47	5.2
	Weak Republican	107	11.8
	Strong Republican	94	10.3
Political Ideology (7-point)			
	Extremely Liberal	165	18.1
	Liberal	242	26.6
	Slightly Liberal	109	12.0
	Moderate	155	17.0
	Slightly Conservative	92	10.1
	Conservative	106	11.6
	Extremely Conservative	41	4.5
Treatment Assignment			
	Treatment (expert info)	482	53.0
	Control	428	47.0
Age (years)			
	Mean (SD)	39.5 (10.6)	
	Range	18–65	

Notes: Sample recruited via CloudResearch Connect. Party identification uses standard 7-point branching format. Treatment assignment conducted via block randomisation by occupation category.

B.2 Descriptive Statistics

Table 6: Descriptive Statistics for Key Variables ($N = 910$)

Variable	N	Mean	SD	Min	Median	Max
Panel A: Outcome and Treatment						
Policy Demand (mean, 1–5)	910	5.49	1.15	1.00	5.60	7.00
Policy Demand (std.)	910	-0.00	1.00	-3.91	0.09	1.31
Treatment Assignment	910	0.53	0.50	0.00	1.00	1.00
Panel B: Psychological Scales						
Trait Anxiety (mean, 1–7)	910	3.65	1.60	1.00	3.75	7.00
Automation Anxiety (mean, 1–7)	910	3.29	1.67	1.00	3.00	7.00
Automation Impact F1 (mean, 1–5)	910	3.11	1.07	1.00	3.25	5.00
Automation Impact F2 (mean, 1–5)	910	3.19	0.94	1.00	3.33	5.00
Economic Anxiety (index, 1–5)	910	2.79	0.93	1.00	2.67	5.00
Locus of Control (index, 1–5)	910	3.44	0.70	1.00	3.50	5.00
Numeracy (correct / 3)	910	2.48	0.74	0.00	3.00	3.00
Panel C: Treatment-Specific Variables[a]						
Trust in Expert Source (1–5)	482	3.13	0.95	1.00	3.00	5.00
Subjective Risk (						
Prior Uncertainty (σ)	468	11.8	7.4	1.0	10.0	40.0
Belief Gap (raw)	482	2.57	36.00	-82.85	2.06	94.16
Belief Gap (std.)	482	0.00	1.00	-2.37	-0.01	2.54
Panel D: Demographic Covariates						
Age (years)	910	39.5	10.6	18.0	38.5	65.0
Female	910	0.52	0.50	0.00	1.00	1.00
White	910	0.69	0.46	0.00	1.00	1.00
Education (1–5)	910	3.68	0.91	1.00	4.00	5.00
Ideology (1–7)	910	3.27	1.81	1.00	3.00	7.00
Party ID (1–7)	910	3.14	2.12	1.00	2.00	7.00
Religiosity (1–6)	910	4.74	1.60	1.00	5.50	6.00
Panel E: Objective Risk						
Automation Risk Percentile	910	41.6	30.7	0.5	32.2	100.0

Notes: Policy Demand measured on 5 items (1–7 Likert, sum range 5–35). Automation Anxiety: 4-item scale, item 4 reverse-coded. Automation Impact: two-factor scale; F1 captures negative personal impacts (4 items), F2 captures positive personal impacts (3 items). Economic Anxiety: mean of job worry, financial worry, and reverse-coded economic hope (1–5). Locus of Control: 4-item internal subscale, items 3–4 reverse-coded (1–5). Uncertainty recoded from ordinal categories to upper-bound positions (1, 5, 10, …, 40); the open-ended top category ("more than 40 spots") is excluded from the mean ($n = 482$ less 14 excluded).

[a] Measured only in the treatment condition ($n = 482$).

B.3 Scale Reliability and Validity

Table 7: Psychometric Properties of Multi-Item Scales ($N = 910$)

Scale	k	α	ω	CFI	TLI	RMSEA	SRMR	Verdict
Policy Demand (Q96)	5	0.824	0.815	0.983	0.943	0.102	0.027	good fit[b]
Trait Anxiety (Q23)	4	0.914	0.911	0.991	0.944	0.163	0.014	good fit[b]
Automation Anxiety (Q94)	4	0.931	0.921	0.997	0.980	0.110	0.006	good fit[b]
Automation Impact (Q95) — 2-factor	7	—	—	0.988	0.974	0.070	0.027	good fit
Locus of Control (Q81)	4	0.475	0.441	0.896	0.375	0.182	0.042	poor fit[b]
Economic Anxiety (Q24/Q25/Q26)	3	0.705	0.727	1.000	—	—	0.000	good fit
Joint Measurement Model (6 factors, LoC excluded)								
23 items, 6 factors	23	—	—	0.958	0.949	0.054	0.048	good fit

Notes: α = Cronbach's alpha; ω = McDonald's omega (from CFA standardised loadings); CFI = Comparative Fit Index; TLI = Tucker-Lewis Index; RMSEA = Root Mean Square Error of Approximation; SRMR = Standardised Root Mean Square Residual. All CFA models include a priori correlated residuals for item pairs sharing a content sub-theme (see text). Verdict is based on a majority vote among CFI, TLI, and SRMR; χ^2/df and RMSEA are excluded from or down-weighted in the verdict because they are inflated at $N > 300$ (Hu & Bentler 1999; Kline 2016). Thresholds: CFI, TLI ≥ 0.95 (good), ≥ 0.90 (acceptable); SRMR ≤ 0.08 (good), ≤ 0.10 (acceptable). LoC is excluded from the joint model because its individual CFA shows poor fit, indicating the 4-item battery is not unidimensional in this sample.

[a] Saturated (just-identified) model; fit indices are trivially perfect.

[b] RMSEA exceeds the 0.08 threshold but is N-sensitive; CFI, TLI, and SRMR support an acceptable or good verdict.

B.4 Scale Inter-Correlations

Table 8: Bivariate Spearman Correlations Among Scale Scores ($N = 910$)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
(1) Policy Demand	—						
(2) Trait Anxiety	0.16***	—					
(3) Automation Anxiety	0.17***	0.21***	—				
(4) Auto Impact F1	0.27***	0.23***	0.72***	—			
(5) Auto Impact F2	−0.23***	−0.16***	−0.12***	−0.35***	—		
(6) Locus of Control	−0.13***	−0.34***	−0.22***	−0.26***	0.19***	—	
(7) Economic Anxiety	0.25***	0.47***	0.38***	0.42***	−0.26***	−0.40***	—

Notes: Spearman rank correlations, listwise complete ($N = 910$). Scale scores: Policy Demand and Trait Anxiety are item means; Automation Anxiety is the mean of four items (item 4 reverse-coded); Automation Impact F1 and F2 are subscale means from the two-factor Q95 model; Locus of Control is the 4-item internal index (items 3–4 reverse-coded); Economic Anxiety is the mean of Q24, Q25, and reverse-coded Q26. Significance: $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

B.5 Trust Mixture Covariates and Epistemic Validation

To probe the *epistemic validation* pattern discussed in the main text, we estimate a trust-mixture measurement model for Q142 in which trust depends on the standardized belief gap (Gap), its square (Gap^2), and a standardized index of prior exposure to automation information (linear and squared terms). Table 9 reports the corresponding γ coefficients.

The results show a weakly negative linear effect of Gap and a clearly negative quadratic term: trust is highest when expert estimates roughly match prior beliefs (Gap ≈ 0) and declines as the discrepancy grows in either direction. Prior exposure exhibits the opposite pattern, with no linear effect but a positive quadratic term, consistent with moderate prior exposure boosting trust but very low or very high exposure having weaker associations. The large positive estimate for δ indicates that the latent Truster and Skeptic classes remain well separated on the Q142 scale even after conditioning on these covariates.

Table 9: Trust Covariate Effects in the Mixture Model (Q142 Measurement)

Covariate	Mean	95% CI (low)	95% CI (high)
Gap (linear)	-0.144	-0.317	0.026
Gap^2 (quadratic)	-0.391	-0.553	-0.243
Prior exposure (linear)	0.043	-0.173	0.272
Prior exposure^2 (quadratic)	0.294	0.026	0.566
Class separation δ	2.085	0.304	3.947

Note: Coefficients are gamma parameters from the trust (Q142) measurement model in the BFMM, with Gap and prior exposure standardized as in the main text.

B.6 Belief Gap: Demographic Subgroups

Figure 6 compares the share of respondents who underestimate their automation risk, mean subjective risk ratings, and mean expert-assigned risk percentiles across demographic subgroups defined by age, party identification, ideology, and education. Wilson confidence intervals are shown for the underestimation share. The underestimation rate hovers near or above 50% in most subgroups, but the source of the gap differs: older workers (>50) and those with a BA or higher underestimate because they hold lower subjective priors despite facing comparable expert risk, whereas Independents and respondents with high-school education or less underestimate because they face higher expert risk that their subjective assessments fail to track. Partisan and ideological differences in the underestimation share are modest, suggesting that the Belief Gap is driven more by occupational sorting into the expert risk distribution than by political identity.

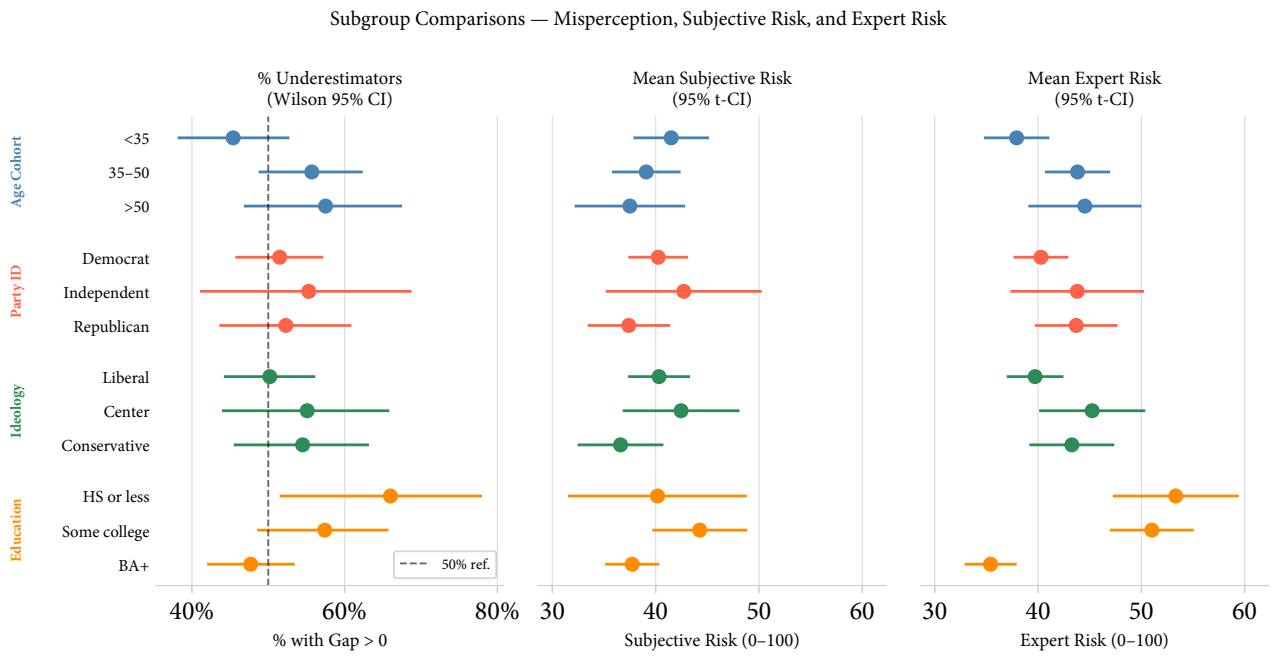


Figure 6: Belief gap by demographic subgroup. Left panel: share of respondents whose subjective risk rating falls below the expert estimate (with Wilson 95% CIs). Center: mean subjective risk. Right: mean expert risk percentile.

B.7 Belief Gap: Calibration Across the Expert Risk Distribution

Figure 7 diagnoses where in the expert risk distribution the belief gap concentrates. The top-left panel shows the distribution of expert risk with decile boundaries; the top-right plots subjective against expert risk with a LOWESS smoother and the 45° perfect-calibration line; the bottom panels display actual-versus-perceived risk

and the share of underestimators by expert risk decile. Subjective priors display a pronounced compression pattern: respondents in the lowest deciles overestimate their risk (arrows point downward in the bottom-left panel), while respondents above roughly the 5th decile increasingly underestimate, with the share of underestimators rising monotonically from under 10% in the 1st decile to nearly 100% in the 10th. The LOWESS curve is nearly flat above expert risk ≈ 30, confirming that subjective beliefs regress toward a common mean ($\sim$35–40) regardless of actual expert-assigned risk. This regression-to-the-mean pattern means that the Belief Gap—and thus the "activation potential" at *Gate 2*—is concentrated among workers in objectively high-risk occupations, precisely the subgroup for whom policy demand is most consequential.

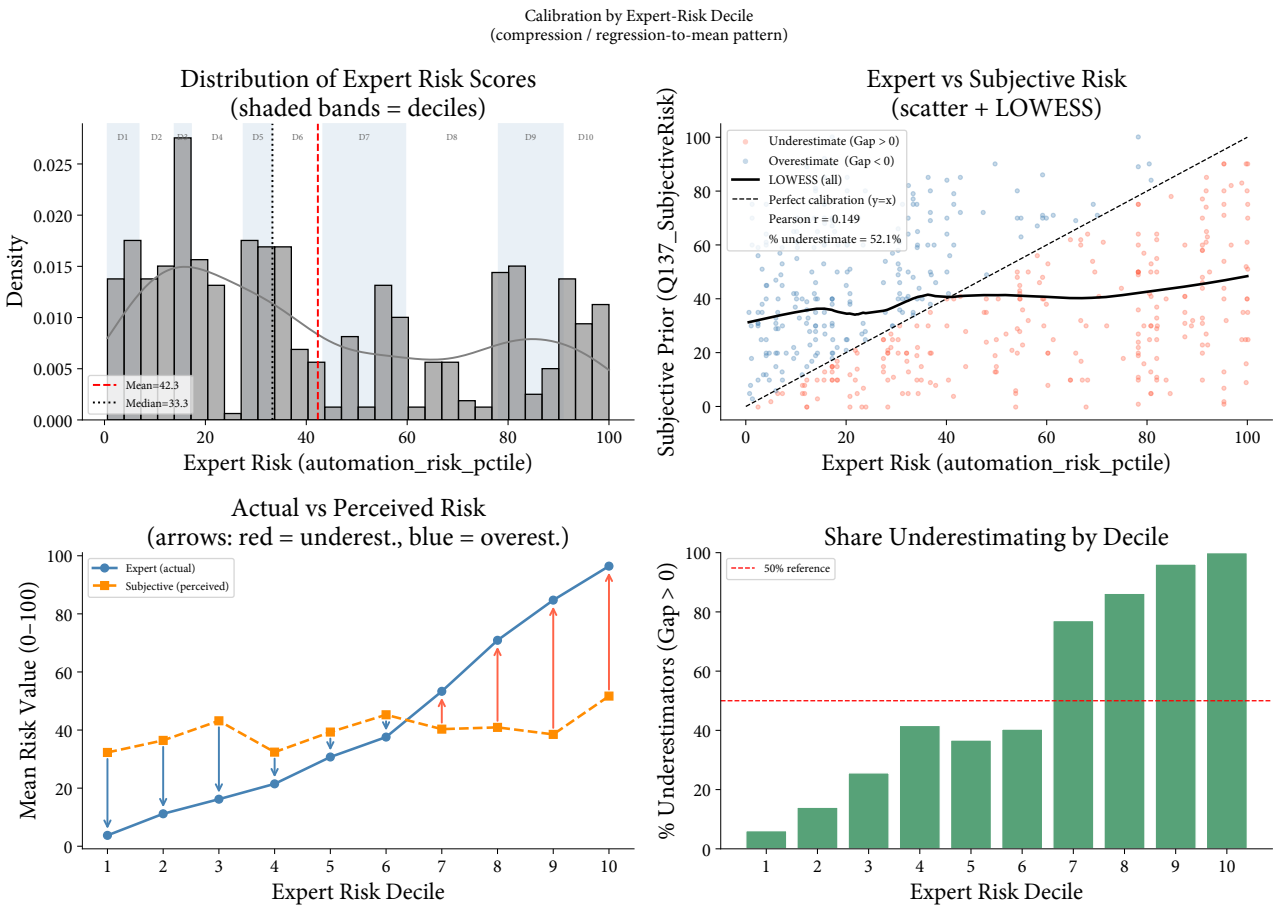


Figure 7: Calibration of subjective risk beliefs across the expert risk distribution. Systematic divergence from the 45° line indicates where respondents most severely underestimate their automation exposure.

B.8 Belief Gap: Certainty vs. Accuracy

Figure 8 contrasts respondents' declared certainty about their automation risk (Q138) against actual accuracy. The left panel shows half-violin and box plots of absolute gap by certainty level with a Spearman ρ and Kruskal-Wallis test; the right panel plots hit rates—whether the actual gap fell within the respondent's self-declared tolerance—with 95% Clopper-Pearson confidence intervals. The left panel reveals no relationship between declared certainty and actual accuracy: mean $|\text{Gap}|$ is virtually constant across all certainty levels (Spearman $\rho = -0.004$, $p = .94$; Kruskal-Wallis $p = .77$), and the distributions overlap almost entirely. The right panel shows that hit rates rise mechanically with wider declared tolerances—from 2% among the most certain (± 1) to 52% among those declaring ± 30—but this reflects the tolerance band widening, not improved accuracy. Respondents who declare themselves most certain are, if anything, the least calibrated: only 2% fall within their own ± 1 margin. This disconnect between metacognitive confidence and actual calibration implies that respondents lack reliable introspective access to the magnitude of their Belief Gap, reinforcing the rationale for using

the externally computed expert–subjective discrepancy rather than self-reported uncertainty as the activation variable at *Gate 2*.

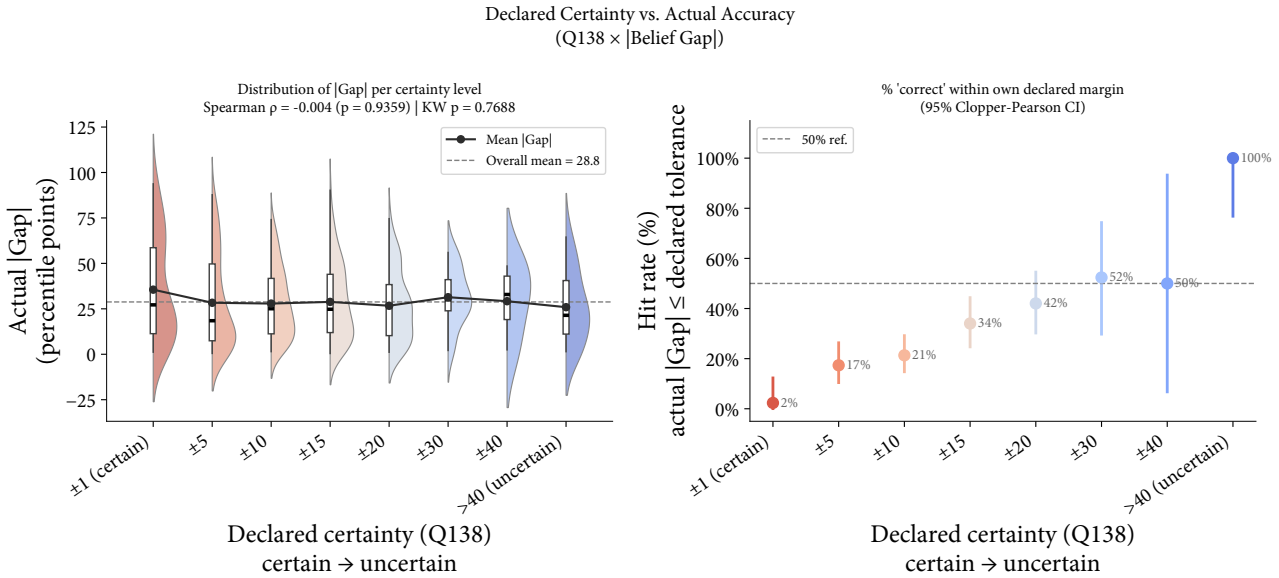


Figure 8: Self-declared certainty vs. actual accuracy. More-certain respondents are not systematically more accurate in estimating their automation risk.

B.9 Trust: Distributional Overview

Figure 9 characterizes the distribution of trust in the expert source among treated respondents and its association with key covariates: the standardized belief gap, expert risk quintile, and prior exposure to automation information. The modal response is "Moderately" (45%), with only about one-third of treated respondents ($n = 162$) reporting high trust (≥ 4). The belief-gap densities (top-right) show that high-trust respondents cluster around small positive gaps, whereas low-trust respondents exhibit a wider, left-shifted distribution with a heavier tail of large negative gaps—consistent with the epistemic validation pattern in which large discrepancies between expert and prior beliefs erode source credibility. Mean trust by belief-gap quintile (bottom-left) is highest in the 2nd and 3rd quintiles and drops in both tails, confirming the inverted-U relationship captured by the Gap^2 term in the measurement model. Prior automation information exposure (bottom-right) rises monotonically with trust: respondents who report the highest trust ($= 5$) have a mean prior-exposure z-score of 0.53, roughly 0.8 SD above the lowest-trust group ($\mu = -0.28$), suggesting that familiarity with the topic bolsters receptivity to the expert source.

B.10 Trust: Partisan and Ideological Associations

Figure 10 cross-tabulates trust with 7-point party identification and 7-point ideology. The top row shows the proportion of each trust level within partisan and ideological categories; the bottom row plots mean trust with 95% confidence intervals across the full scales. The compositional panels (top row) reveal clear partisan sorting: 48% of high-trust respondents are Strong Democrats compared with 27% of the low-trust group, and 35% of high-trust respondents identify as Liberal versus 24% among low-trust respondents. The mean-trust panels (bottom row) show that Strong Democrats report the highest average trust (≈ 3.4) while Independents and Lean Republicans report the lowest (≈ 2.9), a gradient of roughly half a scale point. The ideological pattern is similar but more pronounced at the conservative extreme: Very Conservative respondents average ≈ 2.5, nearly a full point below the overall mean of 3.13. These patterns motivate the partisan sorting concern discussed in the limitations: because left-leaning respondents both trust the expert source more and are predisposed to support government-led policy, trust-moderated treatment effects could partly reflect ideological alignment rather than

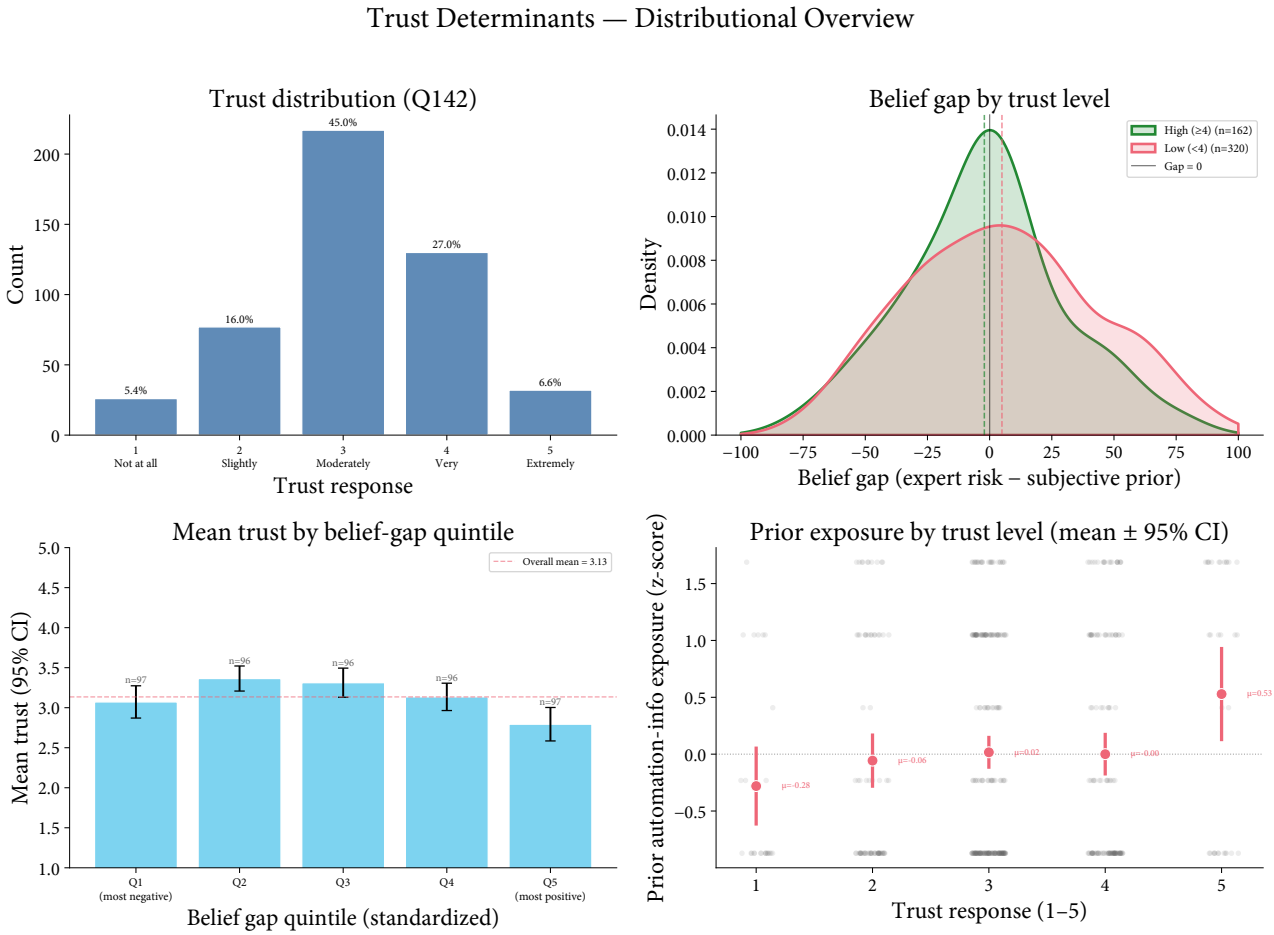


Figure 9: Trust in the expert source among treated respondents ($n = 455$). Top-left: distribution of the 5-point trust item. Top-right: KDE densities of the standardized belief gap by trust level. Bottom-left: mean trust by expert risk quintile (95% CIs). Bottom-right: trust by prior automation information exposure.

dispositional openness to expert information. The structural model addresses this by including ideology and partisanship as controls in the policy demand equation.

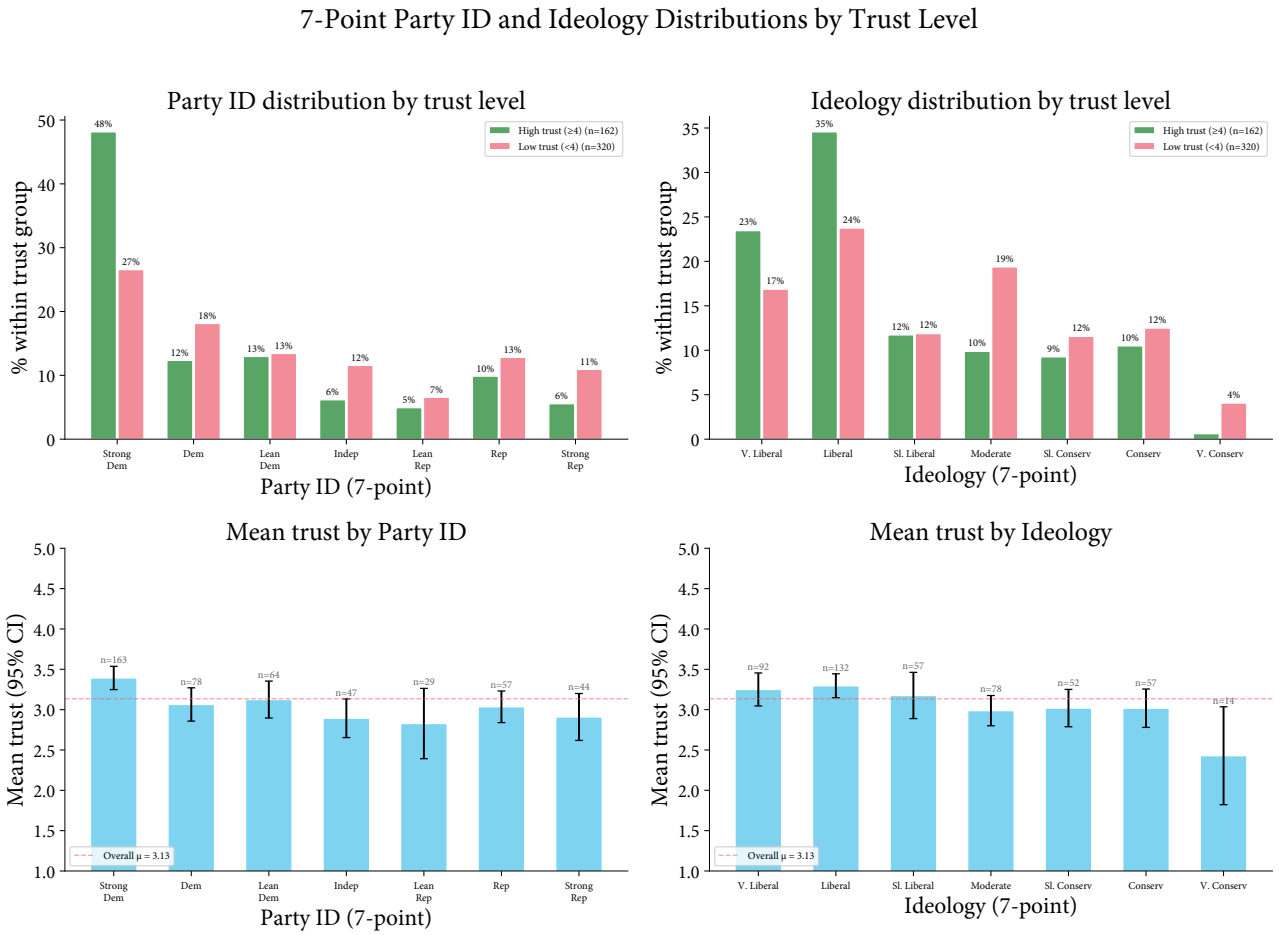


Figure 10: Trust in the expert source by 7-point party identification (left) and 7-point ideology (right). Democrats and liberals report higher trust, consistent with the partisan sorting pattern discussed in the limitations.

C Randomization Balance and Manipulation Checks

This appendix documents the integrity of the experimental design. Section C first verifies that block randomization produced balanced treatment and control groups on all pre-treatment covariates, then presents the full manipulation battery confirming that treated respondents comprehended the information, perceived the expert–subjective belief gap in the correct direction, and updated their beliefs proportionally to that gap.

C.1 Randomization Balance Checks

Tables 10 and 11 report side-by-side means, standardized mean differences, and chi-square diagnostics that verify the block randomization produced comparable treated and control groups along all pre-treatment covariates used in the analysis. Of 13 continuous covariates tested, trait anxiety is nominally significant at $p = .030$ (SMD $= 0.15$) but does not survive Holm correction across the full covariate set (corrected $p = .387$). With 13 simultaneous tests at $\alpha = .05$, one false positive is expected by chance, and the observed effect size is substantively negligible. No covariate differs significantly after correction.

Table 10: Randomization Balance Check — Continuous Covariates

	Treated ($n = 482$)		Control ($n = 428$)			
Covariate	Mean	SD	Mean	SD	Δ	SMD
Age (years)	40.16	10.79	38.79	10.43	1.37	0.13
Ideology (1=very liberal, 7=very conservative)	3.19	1.77	3.36	1.85	-0.17	-0.09
Education (1=less than HS, 5=graduate degree)	3.73	0.91	3.64	0.90	0.09	0.10
Numeracy score (0–3)	2.50	0.71	2.46	0.77	0.04	0.06
Party ID (1=strong Dem, 7=strong Rep)	3.10	2.09	3.19	2.15	-0.09	-0.04
Locus of control — internal index	3.43	0.71	3.44	0.70	-0.01	-0.01
Trait anxiety (mean of 4 items)	3.76	1.61	3.53	1.59	0.23*	0.14
Economic anxiety index (std.)	0.03	1.03	-0.05	0.97	0.08	0.08
Expert automation risk score (0–100)	42.29	30.69	40.81	30.72	1.49	0.05
Female (0/1)	0.52	0.50	0.51	0.50	0.00	0.01
White (0/1)	0.70	0.46	0.68	0.47	0.02	0.05
Risk tolerance index	-0.05	0.98	-0.01	1.01	-0.04	-0.04
Patience index	0.01	0.99	0.01	1.03	-0.01	-0.01

Note: Δ = treated minus control mean. SMD = standardized mean difference (Cohen's d, pooled SD). Stars on Δ indicate significance of two-sided Welch t-test: $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$. Trait anxiety is nominally significant ($p = .030$, SMD $= 0.15$) but does not survive Holm correction across the 13 covariates tested (corrected $p = .387$). With 13 simultaneous tests at $\alpha = .05$, one false positive is expected by chance; the observed SMD is substantively negligible. No covariate differs significantly after correction, confirming successful randomization.

Table 11: Randomization Balance Check — Categorical Covariates

	Treated		Control	
Category	n	%	n	%
Gender [$\chi^2(3) = 0.65, V = 0.027$]				
Male	223	46.3	202	47.2
Female	249	51.7	220	51.4
Other	7	1.5	4	0.9
Prefer not to say	3	0.6	2	0.5
Religious attendance [$\chi^2(5) = 1.48, V = 0.040$]				
Every week	39	8.1	37	8.6
Almost every week	18	3.7	19	4.4
1–2/month	30	6.2	26	6.1
Few/year	83	17.2	79	18.5
Never (religious)	71	14.7	53	12.4
Never (not religious)	241	50.0	214	50.0
Prior info about automation risk [$\chi^2(1) = 0.00, V = 0.002$]				
No	287	59.5	254	59.3
Yes	195	40.5	174	40.7

Note: χ^2 tests for independence between treatment arm and category. V = Cramérś V effect size. Stars: $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

C.2 Manipulation Checks

The manipulation battery verifies three properties: treated respondents attended to the information (comprehension), perceived the expert–subjective gap in the correct direction, and updated their beliefs proportionally to the magnitude of that gap.

C.2.1 Comprehension

A comprehension check after the neutral explanation asked which type of task is easiest for AI to automate. Table 12 shows that 450 of 482 treated respondents (93.4%) selected the correct answer ("routine, clearly defined, repeated patterns"), confirming broad engagement with the treatment text.

Table 12: Manipulation Check: Comprehension (Treated Group Only)

Answer	n	%
Advanced programming skills / technical expertise	17	3.5
Empathy and social connection	4	0.8
Complex simulations / advanced research	11	2.3
Routine, clearly defined, repeated patterns *(correct)*	450	93.4
Total	**482**	**100.0**

Note: 450 of 482 treated respondents (93.4%) selected the correct answer. The comprehension check asked which type of task is easiest for AI to automate.

C.2.2 Perceived vs. Objective Gap

Respondents who received a larger positive gap (expert score exceeding their own estimate) should have perceived the expert assessment as placing their job at higher relative risk. Table 13 confirms a strong monotone relationship: mean Gap rises steadily from -30.9 (those who perceived experts as placing them at significantly lower risk) to $+33.5$ (those who perceived significantly higher risk), yielding a Spearman correlation of $\rho = -0.733$ ($p < .001$) between the perceived direction code and the objective Gap.

Table 13: Manipulation Check: Perceived Gap Direction by Expert–Respondent Belief Gap (Treated Only)

Perceived direction (Q141)	n	Mean Gap	SD Gap
Experts: significantly higher risk	143	35.4	31.9
Experts: slightly higher risk	98	6.2	22.0
Fairly similar risk	71	-2.0	24.4
Experts: slightly lower risk	73	-14.4	17.7
Experts: significantly lower risk	97	-33.4	25.7
Spearman $\rho = -0.733 * ** (p = 0.000)$			

Note: Gap = expert automation score minus respondent subjective score (positive = experts assessed higher risk). Q141 asks respondents whether they perceived experts as placing their job at higher or lower relative risk; code 1 = significantly higher, 5 = significantly lower. Spearman ρ tests whether perceived direction tracks the objective gap. Stars: $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

C.2.3 Belief Updating

Respondents exposed to a positive gap—where expert scores exceeded their own estimates—should report updating their perceived risk upward. Table 14 shows that belief updating tracks the objective gap monotonically: mean Gap rises from -28.1 among those who felt their job was now at much lower risk to $+34.5$ among those who felt it was at much higher risk ($\rho = 0.656$, $p < .001$). Together, the perceived-gap and belief-updating

correlations confirm that treated respondents both correctly interpreted the direction of the expert–subjective discrepancy and revised their risk perceptions accordingly.

Table 14: Manipulation Check: Belief Updating by Expert–Respondent Belief Gap (Treated Only)

Belief update (Q143)	n	Mean Gap	SD Gap
Much lower than I originally thought	55	-35.6	15.9
A little lower than I originally thought	78	-23.5	20.5
About the same as I originally thought	172	3.4	33.6
A little higher than I originally thought	93	15.6	24.1
Much higher than I originally thought	84	35.6	33.0
Spearman $\rho = 0.656 * ** (p = 0.000)$			
Mean Q143: 3.15 (SD=1.22), $n = 482$			

Note: Q143 asks whether respondents now feel their job is at higher or lower risk compared to their original view (1 = much lower, 3 = same, 5 = much higher). Gap = expert score minus respondent subjective score; positive Gap means experts placed the job at higher risk. Spearman ρ tests whether belief updating tracks the objective gap. Stars: $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

C.2.4 Predictors of Baseline Belief Gap

Table 15: Predictors of Expert–Respondent Belief Gap (Treated Group Only)

Predictor	$\hat{\beta}$	SE (HC3)	p
Ideology (std.)	0.415	1.747	0.812
Education (std.)	-6.383***	1.539	0.000
Age (std.)	3.823*	1.717	0.026
Female	1.665	3.408	0.625
White	0.933	3.433	0.786
Numeracy (std.)	2.482	2.214	0.263
Economic anxiety (std.)	-2.528	1.936	0.192
Trait anxiety (std.)	-1.030	2.009	0.608
Locus of control (std.)	3.274	1.883	0.083
Prior automation info (std.)	-7.367***	1.598	0.000
Constant	0.666	3.191	0.835

Note: OLS regression, $n = 482$. $R^2 = 0.129$, adjusted $R^2 = 0.110$. $F(10, 471) = 6.98$, $p = 0.000$. SE = heteroskedasticity-robust (HC3). Dependent variable: belief gap (expert score − respondent subjective score). All continuous predictors standardized. Stars: $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

C.2.5 Treatment Balance Verification

D Exploratory Mediation Analysis

This appendix presents full results tables for the exploratory path analysis of affective mediators discussed in the main text.

Table 16: Balance Verification Summary: Standardized Mean Differences

Covariate	SMD	p-value
Ideology	-0.095	0.154
Education	0.102	0.124
Age	0.129	0.052
Female	0.005	0.938
White	0.046	0.488
Numeracy	0.057	0.396
Trait anxiety	0.145	0.030*
Economic anxiety	0.077	0.247
Expert automation risk	0.048	0.466
Locus of control	-0.011	0.872

Note: SMD = standardized mean difference (Cohen's d). 7 of 10 covariates have $|\text{SMD}| < 0.10$, indicating excellent balance. p-values from two-sided Welch t-tests. Stars: $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

D.1 Treatment Effects on Mechanism Scales

Table 17: Treatment Effects on Attitudinal Mechanisms ($N = 910$)

Outcome	β	SE	p	90% CI	95% CI
Panel A: Mechanism Scales					
Automation Anxiety (Q94)	0.167*	0.066	.012	[0.06, 0.28]	[0.04, 0.30]
Egotropic Threat (Q95)	0.248***	0.066	<.001	[0.14, 0.36]	[0.12, 0.38]
Egotropic Opportunity (Q95)	−0.054	0.066	.418	[-0.16, 0.06]	[-0.18, 0.08]
Panel B: Primary Outcome (for comparison)					
Policy Demand (Q96)	0.275***	0.066	<.001	[0.17, 0.38]	[0.14, 0.40]

Notes: OLS regression with HC3 robust standard errors. All outcomes standardized (mean = 0, SD = 1). β represents the treatment effect in standard deviation units.

Automation Anxiety: 4-item scale measuring job automation concerns (Q94_1–4, with Q94_4 reverse-coded). Egotropic Threat: 4-item scale measuring perceived negative AI career impacts (Q95_1, Q95_3, Q95_5, Q95_7). Egotropic Opportunity: 3-item scale measuring perceived positive AI career impacts (Q95_2, Q95_4, Q95_6).

* $p < .05$; ** $p < .01$; *** $p < .001$.

D.2 Item-Level Treatment Effects

Table 18: Treatment Effects on Automation Anxiety Items (Q94)

Item	Control	Treatment	β	SE	p	90% CI
Q94_1: Job automated in 5 years	2.84	3.00	0.08	0.07	.205	[-0.02, 0.19]
Q94_2: Job automated in 10 years	3.26	3.59	0.16*	0.07	.015	[0.05, 0.27]
Q94_3: Concerned about losing job	3.03	3.41	0.17**	0.06	.007	[0.07, 0.28]
Q94_4: Feel secure (reversed)[a]	3.44	3.69	0.12†	0.07	.077	[0.01, 0.22]

Notes: Structural Equation Model with correlated residuals within scale and full covariate vector (ideology, party ID, age, gender, race, education, religiosity, locus of control, numeracy, trait anxiety). Items measured on 1–7 scale (Strongly Disagree to Strongly Agree). Control and Treatment columns show group means. β is the standardized treatment effect. $N = 910$.

[a] Q94_4 ("I feel secure that my occupation is safe from automation") is reverse-coded; higher values indicate *less* security.

† $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$.

Table 19: Treatment Effects on Perceived AI Career Impact Items (Q95)

Item	Control	Treatment	β	SE	p	90% CI
Threat Items (items 1, 3, 5, 7)						
Q95_1: Reduced job opportunities	3.15	3.41	0.21**	0.06	.001	[0.10, 0.31]
Q95_3: Lower salary	2.95	3.18	0.20**	0.06	.002	[0.09, 0.30]
Q95_5: Decreased job security	2.96	3.27	0.23***	0.06	.000	[0.13, 0.34]
Q95_7: Worsened conditions	2.82	3.08	0.21***	0.06	.001	[0.11, 0.32]
Opportunity Items (items 2, 4, 6)						
Q95_2: New job opportunities	2.88	2.88	0.03	0.06	.651	[-0.08, 0.14]
Q95_4: Enhanced satisfaction	3.16	3.07	-0.05	0.06	.428	[-0.16, 0.05]
Q95_6: Enhanced productivity	3.61	3.54	-0.07	0.07	.271	[-0.18, 0.04]

Notes: Structural Equation Model with correlated residuals within subscale and full covariate vector (ideology, party ID, age, gender, race, education, religiosity, locus of control, numeracy, trait anxiety). Items measured on 1–5 scale. Control and Treatment columns show group means. β is the standardized treatment effect. $N = 910$.
† $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$.

Figures 11–13 visualize the item-level treatment–control comparisons for the three outcome batteries. The treatment shifts negative-valence items (automation timeline beliefs, threat perceptions, and redistributive policy support) while leaving positive-valence and reverse-coded items largely unchanged, consistent with a threat-activation channel rather than a blanket attitude shift. Significance is based on two-proportion z-tests; error bars are 95% Wald confidence intervals.

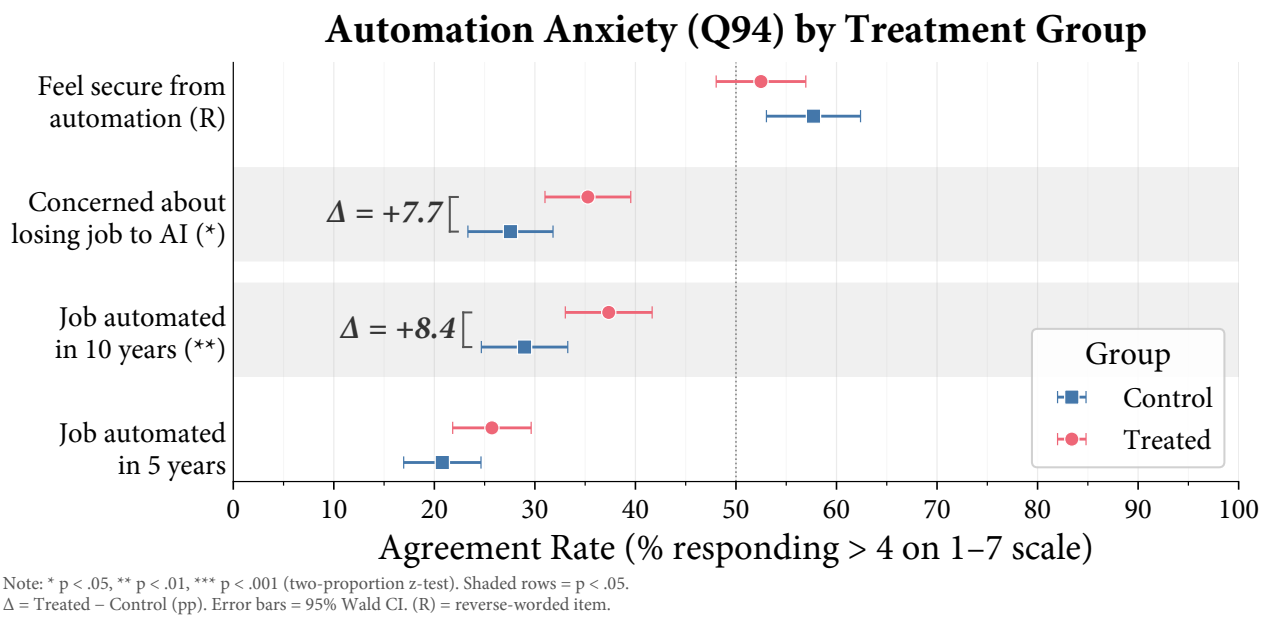
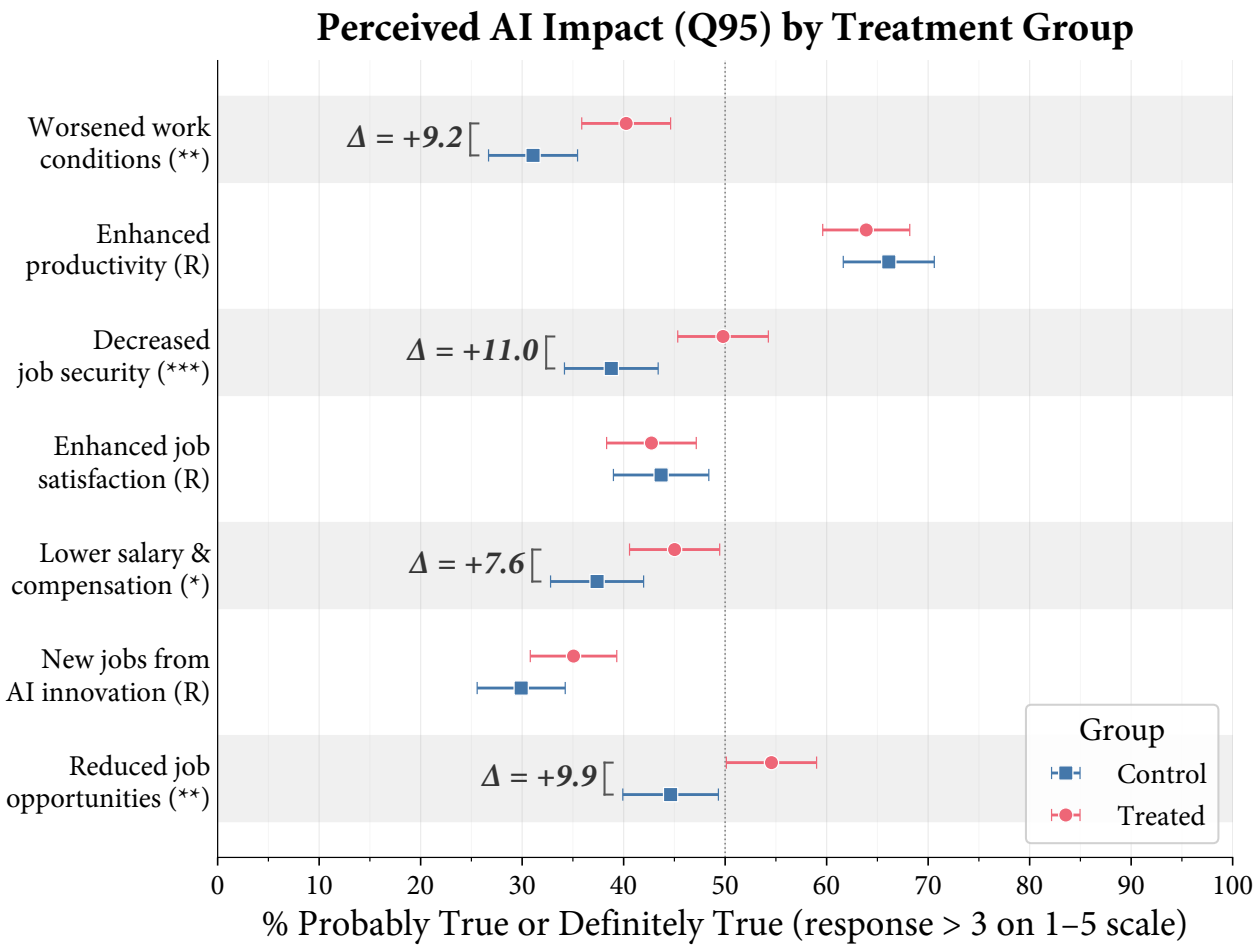
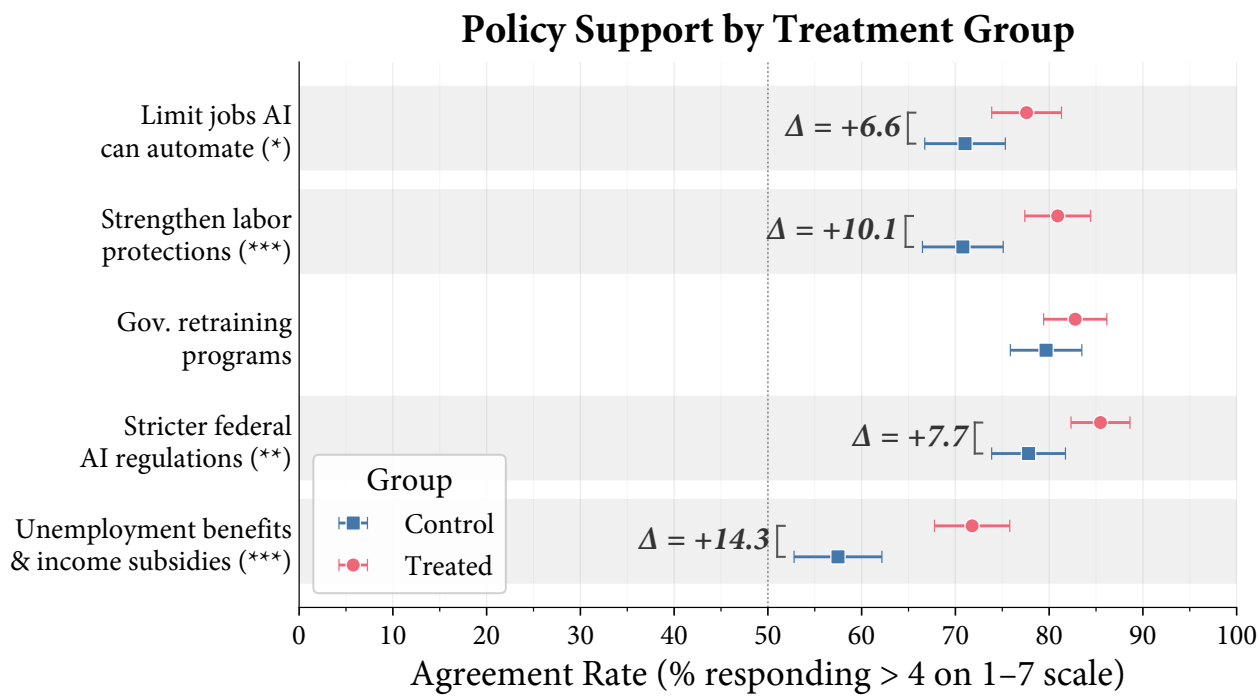


Figure 11: **Automation Anxiety (Q94) by Treatment Group.** Treatment increases agreement on automation timeline items (+8.4 pp for 10-year horizon, +7.7 pp for concern about job loss) but does not significantly shift the reverse-coded security item.



Note: * p < .05, ** p < .01, *** p < .001 (two-proportion z-test). Shaded rows = p < .05.
Δ = Treated − Control (pp). Error bars = 95% Wald CI. (R) = reverse-worded item.

Figure 12: **Perceived AI Impact (Q95) by Treatment Group.** Treatment raises negative impact perceptions (reduced opportunities, decreased security, worsened conditions, lower salary) by +7.6 to +11.0 pp while positive-valence items (productivity, satisfaction, new jobs) remain statistically indistinguishable across groups.



Note: * p < .05, ** p < .01, *** p < .001 (two-proportion z-test). Shaded rows = p < .05.
Δ = Treated − Control (pp). Error bars = 95% Wald CI.

Figure 13: **Policy Support (Q96) by Treatment Group.** Four of five policy items show significant treatment effects, with the largest shifts on redistributive measures (unemployment benefits: +14.3 pp; labor protections: +10.1 pp). Government retraining—the highest-baseline item—is non-significant, consistent with a ceiling effect.

D.3 Mediation Path Decomposition

Table 20: Mediation Analysis: Path Coefficients and Indirect Effects

Path	Coefficient	SE	p	90% CI	95% CI
Panel A: Treatment → Mediator (a paths)					
Treatment → AutoAnxiety	0.15*	0.07	.023	[0.04, 0.26]	[0.02, 0.28]
Treatment → EgoThreat	0.24***	0.06	<.001	[0.13, 0.34]	[0.11, 0.36]
Treatment → EgoOpp	−0.04	0.06	.570	[-0.14, 0.07]	[-0.16, 0.09]
Panel B: Mediator → Policy Outcome (b paths, controlling for Treatment)					
AutoAnxiety → Policy	0.04	0.04	.318	[-0.03, 0.11]	[-0.04, 0.12]
EgoThreat → Policy	0.11*	0.05	.011	[0.04, 0.19]	[0.03, 0.20]
EgoOpp → Policy	−0.08*	0.04	.032	[-0.14, -0.02]	[-0.15, -0.01]
Panel C: Direct and Indirect Effects					
Total Effect (c path)	0.23***	0.06	<.001	[0.14, 0.33]	[0.12, 0.35]
Direct Effect (c' path)	0.20***	0.06	<.001	[0.10, 0.29]	[0.08, 0.31]
Indirect Effects (a × b):					
via AutoAnxiety	0.006	—	—	—	—
via EgoThreat	0.027	—	—	—	—
via EgoOpp	0.003	—	—	—	—
Total Indirect	0.036	—	—	—	—
Proportion Mediated	15.5%				

Notes: All coefficients are standardized (both predictors and outcomes). Standard errors are HC3 robust. Panel A shows the effect of treatment on each potential mediator ("first stage"). Panel B shows the effect of each mediator on policy support controlling for treatment ("second stage"). Panel C decomposes the total treatment effect into direct and indirect components. $N = 910$.
Indirect effects calculated as the product of a and b paths. Bootstrap confidence intervals for indirect effects not reported due to identification concerns (see text).
* $p < .05$; ** $p < .01$; *** $p < .001$.

E Partisan Heterogeneity in Treatment Effects

We examine whether the heterogeneity documented in the Conditional Activation Model could simply reflect partisan predispositions toward government intervention rather than source trust. The appendix text below summarizes the analytic strategy and main conclusions; Tables 21 and 22 provide the full numeric results.

Table 21: Partisan × Trust Mixture CATE Estimates (Exploratory)

Group	ATE (SD)	90% CI	CATEhigh	P(>0.1)
Panel A: Marginal Partisan Effects (ignoring trust class)				
Non-Democrat	0.20	[0.05, 0.34]	—	87%
Democrat	0.35	[0.23, 0.48]	—	100%
Difference (Dem − Non-Dem)	0.15	[-0.04, 0.34]	—	68%
Non-Liberal	0.21	[0.07, 0.34]	—	92%
Liberal	0.35	[0.22, 0.47]	—	100%
Difference (Lib − Non-Lib)	0.14	[0.02, 0.29]	—	67%
Panel B: Cell-Specific Effects (Trust Class × Party)				
Skeptic × Non-Democrat	0.14	[-0.03, 0.30]	-0.21 [-0.56, 0.10]	66%
Skeptic × Democrat	0.29	[0.12, 0.46]	0.31 [-0.05, 0.63]	97%
Truster × Non-Democrat	0.32	[0.12, 0.54]	0.55 [0.12, 1.04]	96%
Truster × Democrat	0.47	[0.17, 0.82]	0.92 [0.28, 1.67]	97%
Panel C: Decomposition (Main Effects and Interaction)				
Trust Main (Truster − Skeptic)	0.18	[-0.07, 0.47]	—	68%
Partisan Main (Dem − Non-Dem)	0.15	[-0.03, 0.34]	—	68%
Trust × Partisan Interaction	0.01	[-0.32, 0.33]	—	≈50%
Panel D: Within-Class Partisan Contrasts				
Within Skeptics (Dem − Non-Dem)	0.15	[-0.09, 0.39]	—	—
Within Trusters (Dem − Non-Dem)	0.16	[-0.23, 0.54]	—	—

Note: Exploratory analysis ($N = 901$). CATEhigh denotes conditional average treatment effect at high Belief Gap (+1 SD) and high trait anxiety (+1 SD), with 90% CI in brackets. Democrat defined as Party ID ≤ 3 on 7-point scale; Liberal defined as ideology ≤ 3 on 7-point scale. Mixture model parameters: $\hat{\pi} = 0.32$ [0.10, 0.62], $\hat{\delta} = 2.26$ [0.37, 4.08]. P(>0.1) denotes posterior probability that effect exceeds 0.1 SD. The near-zero interaction (Panel C) suggests partisan and trust-based heterogeneity operate additively. Within-class contrasts (Panel D) use conservative CI assuming independence of cell ATEs. All chains converged ($\hat{R} < 1.01$, no divergences).

Table 22: Summary of Partisan Heterogeneity Tests vs. Preregistered Criteria

Effect	Est.	HDI $\neq 0$?	P(>0.05)	Decision
Pass 1: Marginal Effects				
Democrat − Non-Democrat	+0.15 SD	Marginal	82%	Suggestive
Liberal − Non-Liberal	+0.14 SD	Yes	85%	Suggestive
Pass 2–3: Interaction Model				
Trust Main Effect	+0.18 SD	No	78%	Suggestive
Partisan Main Effect	+0.15 SD	No	82%	Suggestive
Trust × Partisan	+0.01 SD	No	41%	**Null**
Within-Class Contrasts				
Within Skeptics	+0.15 SD	No	—	Suggestive
Within Trusters	+0.16 SD	No	—	Suggestive

Note: "Suggestive" = consistent positive direction but fails preregistered criteria (90% HDI excludes zero *and* P(>0.05)> 0.95). "Null" = no directional evidence. P(>0.05) computed from posterior normal approximation.

E.1 Full-Sample ATE Robustness and Partisan Interaction Tests

As a further robustness check, the ATE robustness table (available in the replication archive) reports three estimators on the full analytic sample ($N = 910$): (1) unweighted OLS with pre-treatment covariate controls and

HC3 standard errors; (2) WLS using ANES post-stratification weights to reweight the sample to match ANES 2020–2024 partisan proportions (Dem 34.7%, Ind 34.0%, Rep 31.3%); and (3) OLS and WLS interaction models that allow the treatment effect to vary by three-category party ID, with Independents as the base reference. All four specifications yield consistent positive ATEs on standardized policy demand ranging from $+0.21$ to $+0.23$ (all $p \leq .001$), confirming that the main effect is not an artifact of sample overrepresentation of Democrats.

The partisan interaction models produce stratum-specific ATEs of $+0.29$ for Democrats, $+0.15$ for Independents, and $+0.21$ for Republicans (OLS estimates; WLS estimates are nearly identical). None of the interaction contrasts reach conventional significance ($p = .30$ for Dem vs. Ind; $p = .75$ for Rep vs. Ind).

Power caveat. These null interaction results should not be interpreted as evidence of homogeneous treatment effects across partisan groups. Detecting moderation of treatment effects requires roughly four times the sample size needed to detect a main effect of equal magnitude (**?**). With stratum sizes of $N_{\text{Dem}} = 467$, $N_{\text{Ind}} = 242$, and $N_{\text{Rep}} = 201$, the minimum detectable difference between any two stratum ATEs at 80% power is approximately 0.39 SD (Dem vs. Ind) and 0.53 SD (Rep vs. Ind)—far larger than the observed differences of 0.14 SD and 0.06 SD, respectively. The present design is substantially underpowered to detect plausible partisan moderation in the range of 0.10–0.20 SD. Future studies should be designed to target these interaction effects with adequate power, or rely on pooled multi-study meta-analyses.

F Survey Instrument

This appendix reproduces the complete survey instrument, including all question blocks, the experimental treatment stimuli, and outcome measures. Item numbers correspond to the Qualtrics survey flow.

F.1 Survey Flow Overview

The survey followed this sequence:

1. **Demographics & Screening**: State of residence, occupation selection, attention check.
2. **Political & Personal Background**: Party identification, ideology, trait anxiety, economic anxiety.
3. **Numeracy**: Three-item numeracy battery.
4. **Risk, Time, Locus of Control**: Staircase elicitation tasks (adapted from Falk et al.), locus of control items, prior knowledge of automation.
5. **Experimental Section** (Treatment group only):
   - Information on AI and automation
   - Subjective belief elicitation (automation risk percentile)
   - Expert assessment reveal (Frey & Osborne scores)
   - Manipulation and comprehension checks
6. **Outcome Measures**: Job insecurity beliefs, perceived AI impacts, policy preferences.
7. **Final Demographics**: Gender, race/ethnicity, age, education, religiosity.

F.2 Demographics and Screening

Q176: In which state do you reside? [Dropdown]

Q177: Please pick the major occupation group that best matches the main kind of work you do. [Dropdown: SOC major groups]

Q178: Based on your selected major group, please pick the specific occupation that best matches your work. [Conditional dropdown]

Attention Check (Q181–Q182): "So that we know you are paying attention, please ignore the question on screen and select 'Often.'" [Followed by decoy question about dining out]

F.3 Political and Personal Background

Q19: Generally speaking, do you usually think of yourself as a Democrat, a Republican, an independent, or what?

- Democrat (1)
- Republican (2)
- Independent (3)
- Other party (4)

Q20 [If Democrat/Republican]: Would you call yourself a strong [Party] or not very strong [Party]?

Q21 [If Independent]: Do you think of yourself as closer to the Republican Party or to the Democratic Party?

Q22: When it comes to politics, do you usually think of yourself as…

- Extremely Liberal (1) …Extremely Conservative (7)

Q23 (Trait Anxiety—Big Five Short Form):

Here are characteristics that may or may not apply to you. Please indicate the extent to which you agree or disagree with each statement. [7-point scale: Strongly disagree to Strongly agree]

1. I see myself as anxious, easily upset.
2. I feel threatened easily.
3. I worry about things.
4. I am afraid of many things.

Q24–Q26 (Economic Anxiety):

- Q24: How worried are you about losing your job in the near future? [5-point scale: Extremely worried to Not at all worried]
- Q25: So far as you and your family are concerned, how worried are you about your current financial situation? [5-point scale: Extremely worried to Not at all worried]
- Q26: Thinking about your personal economic well-being, how hopeful do you feel? [5-point scale: Extremely hopeful to Not at all hopeful]

F.4 Numeracy Battery

Q29: Imagine that we roll a fair, six-sided die 1,000 times. Out of 1,000 rolls, how many times would the die come up even (2, 4, or 6)? [Open text]

Q30: In the BIG BUCKS LOTTERY, the chance of winning a $10.00 prize is 1%. If 1,000 people each buy a single ticket, how many people would win? [Open text]

Q31: In the ACME PUBLISHING SWEEPSTAKES, the chance of winning a car is 1 in 1,000. What percentage of tickets win a car? [Open text, 0–100]

F.5 Risk and Time Preferences

Q150 (Risk Willingness): How willing or unwilling are you to take risks in general? [0–10 scale, where 0 = "Completely unwilling to take risks" and 10 = "Very willing to take risks"]

Q151 (Time Willingness): How willing are you to give up something that is beneficial for you today in order to benefit more from that in the future? [0–10 scale, where 0 = "Completely unwilling" and 10 = "Very willing"]

Staircase Tasks (adapted from Falk et al. for US context):

- **Patience Task**: Five binary choices between receiving $235 today vs. a larger amount in 12 months. Initial future offer: $360; bounds: [$240, $505]. If participant chooses immediate payment (impatient), future amount increases; if participant chooses future payment (patient), future amount decreases. Iterates 5 times to approximate indifference point.
- **Risk Task**: Five binary choices between a lottery (50% chance of $700, 50% chance of $0) vs. a sure payment. Initial sure payment offer: $375; bounds: [$25, $725]. If participant chooses lottery (risk-seeking), sure payment increases; if participant chooses sure payment (risk-averse), sure payment decreases. Iterates 5 times to approximate certainty equivalent.

F.6 Locus of Control

Q81: Please indicate how much the following statements apply to you. [1–5 scale: Does not apply at all to Fully applies]

1. If I work hard, I will succeed.
2. I'm my own boss.
3. Whether at work or in my private life: What I do is mainly determined by others.
4. Fate often gets in the way of my plans.

F.7 Prior Knowledge of Automation

Q148: Before today, had you seen, read, or heard anything about how likely your job is to be automated? [Yes / No / Not sure]

Q149 [If Yes]: How much had you seen, read, or heard about this topic? [A great deal / Some / A little]

Q146 [If Yes]: Where did you get this information? [Select all: Employer, Government, News media, Social media, Podcasts, Internet search, Friends/family, Other]

F.8 Experimental Treatment (Treatment Group Only)

F.8.1 Initial Information

Q86: Please carefully read the following information.

> **Automation and Your Job**
>
> Artificial intelligence (AI) is getting better at doing work people used to do. Computers already answer customer questions, write basic code, and sort medical images. As AI improves, some jobs may be partly—or fully—done by machines, while new kinds of jobs also appear.
>
> **How do experts judge automation risk?**
>
> Experts look at the small tasks that make up a job. When those tasks are routine, clearly defined, and follow repeated patterns—and there are lots of examples for computers to learn from—AI systems can pick them up more easily. The shorter and more repetitive a task is, the easier it is to automate.

Example job	Typical task	AI risk (today)
IT Help Desk Technician	10-minute help IT tickets that follow a script	High—chatbots can handle most tickets
IT Systems Designer	Plan how the company's tech infrastructure will work over the next year	Lower—needs long-term planning and team buy-in

Other things matter too. Jobs are more at risk when they… can be done entirely online, have lots of recorded data for AI to learn from, or offer workers little power to negotiate. They are less at risk when they… require in-person skills or dexterity, rely on human trust or empathy, or are backed by strong unions or licenses.

F.8.2 Comprehension Check

Q136: According to the information provided, what kind of task is easiest for AI to automate?

1. Tasks that require advanced programming skills and high-level technical expertise
2. Tasks that depend on empathy and social connection
3. Tasks that are part of complex simulations and advanced research
4. **Tasks that are routine, clearly defined, and follow repeated patterns** [Correct]

F.8.3 Belief Elicitation

Q137: Imagine every worker in the country standing on one long line, sorted by how likely their job is to be automated. The very front of the line (position 0) belongs to people whose jobs are safer than almost everyone else's, while the very back (position 100) belongs to those whose jobs are at risk sooner than almost everyone else's. Someone standing in the middle of the line (position 50) would have half the number of workers ahead of them and half behind.

Taking a moment to reflect on your own job, where do you feel you would place yourself on that line right now? [Slider: 0–100]

Q138: How certain or uncertain are you in your previous estimate? About how many positions above or below do you feel it might be off from where experts would put your job?

- I'm absolutely certain—No more than 1 spot above or below
- Within 5 spots above or below
- Within 10 spots above or below
- Within 15 spots above or below
- Within 20 spots above or below
- Within 30 spots above or below
- Within 40 spots above or below
- I'm extremely uncertain—More than 40 spots above or below

F.8.4 Expert Assessment Reveal

Q139: Now let's compare your view with what experts think.

Dr. Carl Benedikt Frey and Dr. Michael A. Osborne are researchers at the University of Oxford, where they investigate the impact of AI and automation on the labor market. Here's how Dr. Frey and Dr. Osborne estimated job automation risk: They started with the U.S. government's O*NET

database, which lists the skills and day-to-day tasks for every job. A panel of AI researchers first tagged 70 sample jobs as either "automatable" or "not yet automatable." A computer model then learned the patterns: jobs heavy on routine, rule-based tasks, fine hand-eye work, or data processing score higher; jobs that rely on creative thinking, social insight, or physical dexterity score lower. That model produced a 0-to-100 risk score for each job—the higher the number, the sooner experts think machines could take over most of its tasks. We use that 0-to-100 score as the expert estimate for your occupation.

Q140: [Dynamic bar chart displayed]

Your view vs. the expert estimate

Below, you'll see where you placed your job on the automation risk scale, compared to the expert estimate for your occupation.

You placed yourself at: [SUBJECTIVE SCORE]

Experts place your job at: [OBJECTIVE SCORE]

Both bars use the same 0–100 line you saw earlier (0 = safest, 100 = riskiest).

F.8.5 Post-Treatment Checks

Q141 (Manipulation Check): How does the experts' assessment compare to your initial view?

1. The experts place my job at a significantly higher relative risk
2. The experts place my job at a slightly higher relative risk
3. The experts place my job at a fairly similar relative risk
4. The experts place my job at a slightly lower relative risk
5. The experts place my job at a significantly lower relative risk

Q142 (Trust): To what extent do you think the Oxford researchers are a trustworthy source of information about job automation risk?

1. Not at all trustworthy (1) …Extremely trustworthy (5)

Q143 (Belief Updating): Compared to your original view, do you now feel your job is at higher or lower risk of automation?

1. Much lower than I originally thought (1)
2. A little lower than I originally thought (2)
3. About the same as I originally thought (3)
4. A little higher than I originally thought (4)
5. Much higher than I originally thought (5)

F.9 Outcome Measures

F.9.1 Job Insecurity and Automation Beliefs

Q94: How much do you agree or disagree with the following statements? [1–7: Strongly disagree to Strongly agree]

1. My job is likely to be automated in the next 5 years.
2. My job is likely to be automated in the next 10 years.

3. I am concerned about losing my job to automation or AI.
4. I feel secure that my occupation is safe from automation.

F.9.2 Perceived Impacts of AI

Q95: To what extent do you believe that advances in AI will have the following impacts on your career and job prospects? [1–5: Definitely False to Definitely True]

1. Reduced job opportunities: AI will lead to fewer available jobs for me, making it harder to find a job.
2. New jobs: AI-driven innovation will spur economic growth and create new job opportunities for me.
3. Lower salary and compensation: AI will cause a decline in my future wages and benefits in my field.
4. Enhanced job satisfaction: AI's automation of routine tasks will free up my time for creative and strategic work.
5. Decreased job security: AI will make my employment less stable or increase my risk of being laid off.
6. Enhanced productivity: AI will provide tools and assistance that allow me to complete my work more efficiently and effectively.
7. Worsened work conditions: AI will lead to a decline in my pay, benefits, or the quality of my work environment.

F.9.3 Policy Preferences (Primary Outcome)

Q96: To what extent do you support or oppose the following policy responses to address AI-driven automation's workforce impact? [1–7: Strongly Oppose to Strongly Support]

1. Expanding unemployment benefits and income subsidies.
2. Introducing stricter federal regulations on AI deployment, such as requiring companies to undergo ethical AI reviews.
3. Funding government or private-sector retraining programs to help workers transition to new careers.
4. Strengthening labor protections, such as supporting unionization efforts in AI-exposed industries.
5. Limiting the number of jobs that AI can automate within companies to protect human employment.

F.10 Final Demographics

Q104 (Gender): Male / Female / Other / Prefer not to say

Q105 (Race/Ethnicity): [Select all that apply] White, Black or African American, Hispanic or Latino, Asian, American Indian or Alaska Native, Native Hawaiian or Other Pacific Islander, Other, Prefer not to say

Q106 (Age): [Numeric entry]

Q107 (Education): Less than high school / High school diploma / Some college / Bachelor's degree / Graduate degree

Q108 (Religiosity): Thinking about your life these days, do you ever attend religious services, apart from occasional weddings, baptisms, or funerals?

1. Every week
2. Almost every week
3. Once or twice a month

4. A few times a year
5. Never, but I am religious
6. Never, I am not religious

G Pre-Registered Hypotheses (Verbatim)

The following hypotheses were pre-registered on the Open Science Framework prior to any analysis of the data. We reproduce them verbatim to ensure transparency regarding our confirmatory predictions.

Definitions:

- **Treatment manipulation**: The "Belief Correction" mechanism where participants estimate their own automation risk (0–100) and are then shown a dynamic visualization contrasting their subjective estimate against the objective risk score calculated by Frey and Osborne (2017), accompanied by an explanation of the expert methodology.
- **Policy response vector**: The five specific government interventions measured in Q96: (1) Expanding unemployment benefits and income subsidies, (2) Stricter federal regulations on AI deployment, (3) Funding retraining programs, (4) Strengthening labor protections (unionization), and (5) Limiting the number of jobs AI can automate.
- **"Policy Demand"**: The latent construct or composite score derived from the Policy Vector, representing the subject's overall desire for government intervention to mitigate the workforce risks of AI.

The Conditional Activation Model suggests that looking for simple "Main Effects" is a theoretical error. Therefore, our hypotheses follow the logic of the funnel:

H1: The "Weak Main Effect" Hypothesis
The unconditional Average Treatment Effect (ATE) of the Treatment manipulation (Belief Correction) on Policy Demand will be small or statistically insignificant.

H2: The Trigger Hypothesis (Potential Energy) (Directional)
Treatment effects will be conditional on the Belief Gap. Individuals who discover they are at higher risk than they thought (Positive Shock) will show greater Policy Demand than those whose beliefs are confirmed or reassured.

H3: The Gate 1 Hypothesis (Trust Filter) (Directional)
The effect of the Belief Gap is moderated by Source Credibility.

- **High Trust**: The Gap significantly increases Policy Demand.
- **Low Trust**: The Gap is rendered inert; even a massive risk shock will not change Policy Demand if the source is rejected.

H4: The Gate 2 Hypothesis (Dispositional Filter) (Directional)
Among those who trust the source, the effect is further moderated by Trait Anxiety (a general psychological disposition).

- **High Trait Anxiety**: The objective risk signal resonates strongly, triggering a large increase in Policy Demand (Accuracy Motivation).
- **Low Trait Anxiety**: The signal triggers a baseline increase in Policy Demand, but significantly weaker than in high-anxiety individuals.

H5: The Conditional Activation Hypothesis (Directional)
This is the core test of the theory. The change in Policy Demand (Y) is conditional on surviving both gates:

- The strongest causal effects will be observed in the specific subset of participants who survive both filters: High-Trait-Anxiety Trusters who experience a Positive Belief Gap (i.e., this group will show significantly higher Policy Demand compared to the average participant or those with low Anxiety and low Belief Gap).

Note: The full pre-registration document, including the study design, sampling plan, analysis plan, and inference criteria, is available on the Open Science Framework. The OSF link is currently embargoed to preserve anonymity during peer review and will be made public upon acceptance.

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